Supplementary Materials for
**Linking Trait Items of Self-Control to Broader Conceptualizations in Daily Life Using Machine Learning**

# Table of Contents

**Please note:** The correlations between the selected trait predictors and all outcomes, used to evaluate whether the pattern of associations differed across outcomes, can be found in the "Tables" folder in the repository, in files named correlations_[outcome type].xlsx."

# Deviations from the preregistration

**Table S1**

*Preregistration Deviations Table*

| **Deviations** | | | | | |
|---|---|---|---|---|---|
| # | Details | | Original Wording | Deviation Description | Reader Impact |
| *Deviations during the peer review process* | | | | | |
| 1 | Type | Research Q(s) | "RQ1: Which trait predictors are most predictive for each component of self-control?<br>RQ2: Do the most important trait predictors differ across the three types of conflict?<br>RQ3: How well is each aspect of self-control captured by commonly used self-control scales?" | To allow a closer mapping between the research questions and the goals of the study, we changed the order of the questions (RQ3 is the new RQ1), merged two questions (RQ1 and RQ2 into the new RQ2) and slightly changed the wording:<br>RQ1: How strongly do commonly used trait items of self-control predict the components and types of self-control?<br>RQ2: What are differences in the most important trait predictors between the components and types of self-control conflicts? | We hope that this change enhances the clarity of our study, such that the reader can judge our contribution more easily. It is important to note that this reorganization of our research questions do not change the preregistered analyses, but rather how the paper is being structured and especially how the results are being presented.<br><br>Nevertheless, in the peer-review process, we also made some substantial changes in how we interpret the results (see the following point) |
| | Reason | Peer review | | | |
| | Timing | After results knownRQ3 | | | |

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| 2 | Type | Analysis | This deviations concerns differences in what we count as a meaningful difference between the components and types of self-control.<br><br>Perhaps add one more and say that we changed the robustness check; this could an addititional point | *For RQ1 (previous RQ3), we now plot the full distribution of $R^2$ values across all 100 folds rather than only reporting the average, and only interpreted differences that were large and consistent across folds.*<br><br>*For RQ2 (previous RQ1 and RQ2), we removed interpretations of small differences in the magnitude of correlations between trait predictors and outcomes, whether comparing different predictors' correlations with the same outcome or comparing correlations across different predictor-outcome pairs, as these are most vulnerable to sampling error.. Instead, we focused on (1) the content of the selected predictors and (2) the direction of their bivariate correlations. However, since also these may be bound to sampling variation, we took the following steps. First, we inspected the correlations between each trait predictor and all outcomes regardless of selection status, which can be found in the and* | We deemed this change necessary in order to ensure the robustness of the findings. |
| | Reason | Peer review | | | |
| | Timing | After results known | | | |

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| | | | | *refrained from interpreting associations where the correlational pattern did not look descriptively distinct across outcomes. Second, we prioritized interpretations supported by convergent evidence across multiple items, strong bivariate and partial regression associations, and/or consistency with previous research findings. Third, for associations that did not clearly meet the above criteria but were still substantively interpreted, we conducted 1000 random split-half replications to assess whether the interpreted pattern of associations was consistent across independent subsamples.* | |
| 3 | Type | Analysis | Regarding the comparison in the new RQ2: "If the trait predictors in a model entail single items as well as composite scores, we will conduct a robustness check of the rank-order given that composite scores are more reliable than single items. In that case, we will repeat the analyses using only a single item from each composite score with the highest loading in the EFA to allow a fair comparison. | Given that in the revised version, we refrained from interpreting differences in the magnitude of associations between the selected trait predictors and the outcomes, this preregistered robustness check was removed and replaced with other checks (see above). | Leaving the preregistered robustness check out will not affect the conclusions since these do not depend on the results of this check anymore. |
| | Reason | Peer review | | | |
| | Timing | After results known | | | |

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| | | | Additionally, the R-squared values from all folds will be averaged to provide an overall measure of the model's explanatory power.” | | |
| *Deviations before the peer review process (which refer to the preregistered RQs; cf. deviation #1)* | | | | | |
| 4 | Type | Variables | “Predictor items of scales of related constructs: […] One facet of neuroticism (depression) and one facet of extraversion (excitement seeking) from the IPIP-240 (16 items; Goldberg, 1999; Treiber, 2013) | Due to a small error in the codebook of the larger project, the first author who was primarily responsible for the preregistration thought that these variables were assessed, which was not the case. To stay as close as possible to the preregistration, we instead included the extraversion and neuroticism items from the BFI-K (three per dimension). | This change in the included variables means that the depth with which (facets) of neuroticism and extraversion that are considered particularly relevant for self-control is substantially reduced. As such, it may has diminished the ability of our study to detect nuanced differences in how the outcomes may related differently to those facets. However, since those facets are considered construct that are related to self-control and that there are many constructs which are theoretically related but were not included in the current study, we believe that this change should not affect the conclusions substantially. |
| | Reason | Typo/Error | | | |
| | Timing | After data access | | | |
| 5 | Type | Variables | “Predictor items of scales of related constructs: […] The “BIS/BAS scales (24 items; Carver & White, 1994; Strobel et al., 2001)” | The BIS/BAS scales also entail four ‘filler items’ which are unrelated to the constructs. We forgot to exclude those four items from the pregregistration. | Since those four items did not intend to measure constructs that are relevant for the current study, their exclusion ensures that only items aiming to assess constructs specifically stated in the preregistration are included. |
| | Reason | Typo/Error | | | |
| | Timing | After data access | | | |

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| 6 | Type | Sample | “This resulted in a final sample with N = 492 participants who were on average 24.8 years old (SD = 6.4 years, age range: 18–62 years) and 68.90% female.” | This information was taken from another paper which used the same dataset since the first author who conducted the analysis had not yet accessed the data at the time of the preregistration. However, one participant had to be excluded from the preregistered sample as the first author was unable to match the ESM and baseline data for that participant. | The removal of one participant from the preregistered sample due to the inability of matching the data from different study part is not expected to substantially impact the results of the study. |
| | Reason | Typo/Error | | | |
| | Timing | After data access | | | |
| 7 | Type | Analysis | “In the inner loop, hyperparameter tuning will be conducted using a grid search with a resolution of 5000 […]” | In the preparatory phase of the project, we tested the models on a dataset with similar characteristics and observed small variations in outcomes depending on the hyperparameters, which motivated our preregistered decision to use a grid resolution of 5000. However, when applying this setup across all 12 models in the main analysis, the computational cost proved to bedisproportionately high. We therefore reduced the resolution to 2000, which substantially improved efficiency while yielding nearly identical results in additional test runs. This adjustment also facilitates the | We do not expect this change to substantially impact the conclusions of the study. Although hyperparameter resolution can influence model results, we used a still fine-grained grid, repeated the modeling process 10 times, and focused our interpretation on broader patterns (rather than overinterpreting single items) which en the robustness of our findings despite minor parameter variation. |
| | Reason | Other (Please Explain) | | | |
| | Timing | After results known | | | |

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| | | | | reproducibility of our results. | |
| 8 | Type | Analysis | "The EFA will be estimated using the maximum likelihood, and the results of oblimin and promax rotation will be compared." | We only based the decision on the number of factors on the results of the oblimin rotation and report the results of the promax rotation only for the sake of transparency in the OSF folder. We made this decision to reduce the complexity of the analyses. That is, for most of the 12 models, we ran multiple EFA solutions and compared the results. Doubling the number of EFA solutions that had to be compared by considering oblimin and promax rotations was then not deemed raher unfeasible. | Since the goal of the EFA was exclusively to faciliate the interpretation of the selected items and the outcomes, but not to find replicable factors among different scales, this deviation is not expected to substantially impact the conclusions of the study. |
| | Reason | Typo/Error | | | |
| | Timing | After results known | | | |
| | Reason | Typo/ Error | | | |
| | Timing | After results known | | | |
| 9 | Type | Analysis | "To obtain unbiased coefficients that can be more straightforwardly compared with prior literature, we will estimate the relationship between the selected items using linear regression models. […] The linear regression models will be estimated using a 10-fold CV approach, repeated 10 times, resulting in a total of 100 folds. Coefficients from these models will be averaged across all folds." | We decided to use bivariate correlations instead of partial regressions to compare the strength of the associations between the trait predictors and the outcomes for several reasons. First, partial regression coefficients may not be ideally suited given the unclear causal structure among the final trait predictors which, for example, make it difficult to interpret suppression effects that we | We deemed this change necessary in order to avoid drawing false conclusions. |
| | Reason | Typo/Error | | | |
| | Timing | After results known | | | |

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| | | | | observed in some cases. Second, partial regression coefficients depend on the other predictors in the model, thereby limiting the comparison across models with past literature. We still display the results based on the partial regressions in this document, but obtain the coefficients based on the full sample as this is recommended for interpretational purposes (Pargent et al., 2023). | |
| 10 | Type | Analysis | "For RQ1, the selected trait predictors will be descriptively compared across the components of self-control in daily life (frequency, intensity, and success of self-control conflicts). For the comparison, we will focus on the scale origin, including the subscale or theme (as indicated by the EFA) they belong to if applicable, as well as the ABCD content."<br>"For RQ2, the only difference to RQ1 will be that the results are descriptively compared across the nine aspects of self-control in daily life, meaning that the models for the frequency, intensity, and success of self-control conflicts | After conducting the analyses, we realized that the preregistered inferential comparison of the ABCD content across components and types was flawed (cf next point) and an exclusively descriptive comparison hardly feasible. We thus decided to remove these analyses from the main manuscript and only report them in the supplement material for the sake of transparency. | We initially preregistered the comparisons of the ABCD content of the most important predictors because we hoped that this would enrich the possibilites of finding meaningful theoretical differnces between the components and types of self-control. Thus, removing the comparison from the ABCD content undoubtedly impacted the results. Nevertheless, we believe that with the remaining analyses, we are still able to illustrate that there are meaningful theoretical differences between the components and types of self-control. |
| | Reason | Plan not possible | | | |
| | Timing | After results known | | | |

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| | | | will be estimated separately for each conflict type (inhibition, initiation, persistence)." | | |
| 11 | Type | Analysis | "To account for this, we will compute weighted averages of the ABCD codings for each component, where the coefficients determine the weight with which the ABCD coding of each trait predictor impacts the average of each component." | After applying the weighting procedure as specified in the initial preregistration, we realized that it leads to a situation where the ABCD codings do not add up to 100 anymore, which diminishes the comparability across outcomes. To deal with that, we tried to normalize the values by dividing each weighted rating by the total sum of the weighted ratings and then multiplied by 100. However, given that the coefficients were very small (i.e., ranging from 0.01 to 0.05), the normalization attempt seem to largely nullify the minor differences introduced by such small weights such that there was no visible difference in the plots between the normalized and weighted codings and the 'raw' codings. Afterwards, we exlpored different visualization forms, hoping that we would find a solution that still enables an accurate comparison of the ABCD | Same as above. |
| | Reason | Typo/ Error | | | |
| | Timing | After results known | | | |

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| | | | | codings across predictors, but did not find a satisfying solution. Thus, we decided to remove the ABCD codings from the main manuscript. | |
| 12 | Type | Analysis | "For RQ3, the explained variance of the models of each aspect of self-control in daily life is of concern. […] When descriptively comparing the average R-square of those models across aspects, attention will be paid to the scale origin of the most important trait predictors in each model, given the substantial variation in the frequency of use of different self-control scales in the literature." | We decided to disregard the scale origin in the comparison as different scales capture very similar content (also supported by the factors found in the EFAs) but likely only some of those items out of a group of correlated items was selected as final items due to the regularization applied in the item selection procedure. | We deemed this change necessary in order to avoid drawing false conclusions. However, we acknowledge that the disregard of the scale origin only allows to draw rather broader conclusions, which we tried to consider. Thus, we do not expect that this deviation substantially impacts the two main conclusions from RQ3 which are (a) that items from self-control scales seem to reflect all three components but successful resolution substantially stronger, and (2) that the scales seem to mostly reflect initiaiton conflicts followed by persistence conflicts. |
| | Reason | Typo/ Error | | | |
| | Timing | After results known | | | |

**Unregistered Steps**

| # | Details | | Original Wording | Unregistered Step Description | Reader Impact |
|---|---|---|---|---|---|
| 1 | Type | Variables | We did not plan to change the scoring | After plotting the results for the | This unregistered step led to a switch |

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| | Timing | After results known | of the items in the original preregistration. | first time, we decided to score all items in the same direction as their corresponding scale to facilitate the interpretation of the results. | in the sign of the regression coefficient for predictor variables to which it was applied but did not affect any conclusions. |
| 2 | Type | Data Preparation | “To be included in further analyses, participants had to complete the baseline questionnaire and report at least one conflict.” | It was not specified how missing values would be treated in the original preregistration. Since the ML models require complete data, missing values among the predictor items were replaced with mean imputation that was implemented in the nested resampling schedule. For the outcome variables, no data was imputed, and participants were excluded on an outcome-to-outcome basis. | Due to the very low percentage of missing values among the predictor items, the unregistered mean imputation should have only a very small impact on the results. In contrast, the decision to exclude participants on an outcome-to-outcome basis affects the comparison of the different types of conflicts as those models are based on partly different participants. Since this affects the comparability of the different models, it should also affect the readers’ interpretation of the results. |
| | Timing | After data access | | | |
| 3 | Type | Analysis | “While the primary focus of this step is on item selection based on model coefficients, it's crucial to ensure a reasonable model fit before interpreting these coefficients. For a model to be considered in the subsequent steps, the aggregated | It was not clearly specified whether we would apply this “minimal” performance criterion to the regularized models (i.e., the elastic net models) or to the unregularized models (i.e., the linear regression models). We | This decision ensures that many of the models are not excluded too early from the analyses. Since we still applied the minimal criterion – but to the “correct” model on which we also base the conclusions – this decision was deemed necessary to ensure a fair |
| | Timing | After results known | | | |

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| | | | correlation between the observed and predicted values should at least exceed r > .30.” | decided that it is more important that the unregularized models meet this criterion as the final interpretation of the coefficients is based on these models. This means that even if the correlation between observerd and predicted values in the regularized models was slighlty below .30, we still created three predictor sets and tested them in the unregularized models. Only if none of these three predictor sets met this threshold, we concluded that the predictive performance was too poor to select the most important predictors. | compairson between the components and types of self-control. |
| 4 | Type | Analysis | “If we find that items form a clear and interpretable factor in EFA, we will compute a composite (i.e., average) score for those items for subsequent analysis and only interpret the relationship between this composite score and the outcome (but not between the single items). All items that do not form a clear factor will remain independent in subsequent analyses. The primary objective of this step is to facilitate the interpretation of the relationship between the selected items and the outcomes in subsequent | We only computed for items for which the bivariate correlation with the outcome was in the same direction. In the preregistered analysis, we did not specify that we would consider this correlation for the decision of whether a given item should be aggregated into a composite score with other items or not. | This unregistered analysis step was deemed essential to achieve the preregistered goal of the aggregation of predictor items into composite scores which was to facilitate the interpretation of the relationship between the selected items and the outcomes. |
| | Timing | After results known | | | |

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| | | | stages, and not finding latent variables among the items of various self-control scales." | | |

*Note.* The template for this table was created by Willroth and Atherton (2024). Deviations made in response to the peer-review process are described at a more general level, as they were broader in scope than the other listed deviations; given the extent of these changes, we no longer regard the study as preregistered.

## Identifying the most importat predictors for each outcome

### Frequency across self-control conflicts

In a very first step, all 169 trait items were used in regularized ML models. In these regularized models, the aggregated performance across 100 folds for the prediction of the frequency across conflicts was below the preregistered minimal criterion (MSE = 0.053, $R^2$ = .027, $r$ = .236), which we defined as a correlation between observed and predicted values of at least .30. Yet, the performance of the subsequent unregularized models which used three sets of items that were selected (i.e., had non-zero coefficients) in at least 70, 80, or 90 folds showed a satisfactory performance, which is why we continued with the analyses. Despite small differences between the predictor sets, the 13 items that were selected in 90% of the folds performed slightly better (MSE = 0.048, $R^2$ = .112, $r$ = .389) than the 17 items that were selected in 80% of the folds (MSE = 0.048, $R^2$ = .108, $r$ = .382) and the 19 items that were selected in 70% of the folds (MSE = 0.049, $R^2$ = .097, $r$ = .379). Consequently, 13 items were selected as the most important predictors of the frequency across self-control conflicts.

### Intensity across self-control conflicts

The aggregated performance of the regularized model using all trait items was again slightly below the minimal criterion (MSE= 0.668, $R^2$ = .048, $r$ = .280). However, as with the models for the frequency across self-control conflicts, the performance in the unregularized models using the three predictor sets was satisfactory. Despite again very small difference in the predictor sets, the 20 items that were selected in 90% of the folds (MSE = 0.626, $R^2$ = .101, $r$ = .376) performed very slightly better than the 28 items that were selected in 70% of the folds (MSE = 0.629, $R^2$ = .091, $r$ = .377) and the 24 items that were selected in 80% of the folds (MSE = 0.632, $R^2$ = .090, $r$ = .375). Consequently, 20 items were selected as most important predictors for the intensity across self-control conflicts.

### Success across self-control conflicts

The aggregated performance of the regularized model using all trait items was satisfactory (MSE= 0.835, $R^2$ = .132, $r$ = .408). The 17 items that were selected in 90% of the folds performed better (MSE= 0.771, $R^2$= .198, $r$ = .487) than the 32 items that were selected in 70% of the folds (MSE= 0.784, $R^2$= .185, $r$ = .476) and the 26 items that were selected in 80% of the folds (MSE= 0.791, $R^2$= .180, $r$ = .467). Consequently, 17 items were selected as most important predictors for the success across self-control conflicts.

### Frequency of initiation conflicts

The aggregated performance across 100 folds of the regularized models using all trait predictors was below the minimal performance criterion (MSE = 0.029, $R^2$ = .013, $r$ = .223). However, the 10 items that were selected in 70% of the folds showed a satisfactory performance in the unregularized models (MSE = 0.027, $R^2$ = .066, $r$ = .329), unlike the 7 items that were selected in 90% of the folds (MSE = 0.028, $R^2$ = .039, $r$ = .293) and the 9 items that were selected in 80% of the folds (MSE = 0.028, $R^2$ = .039, $r$ = .285). Consequently, 10 items were selected as most important predictors for the frequency of initiation conflicts.

## Frequency of persistence conflicts

The aggregated performance across 100 folds of the regularized models using all trait predictors was very poor (MSE = 0.012, $R^2$ = -.042, $r$ = .094). However, the 11 items that were selected in 70% of the folds showed a satisfactory performance in the unregularized models (MSE = 0.011, $R^2$ = .033, $r$ = .306), unlike the 5 items that were selected in 80% of the folds (MSE = 0.011, $R^2$ = -.001, $r$ = .231) and the 3 items that were selected in 90% of the folds (MSE = 0.011, $R^2$ = -.012, $r$ = .226). Consequently, 11 items were selected as most important predictors for the frequency of persistence conflicts.

## Frequency of inhibition conflicts

The aggregated performance across 100 folds of the regularized models using all trait predictors was very poor (MSE = 0.008, $R^2$ = -.038, $r$ = .002). In contrast to the other conflict types, the analysis had to stop at this point since no trait predictor was selected in at least 70% of the folds and thus no unregularized models could be run.

## Intensity initiation conflicts

The aggregated performance across 100 folds of the regularized models using all trait predictors was below the minimal performance criterion (MSE = 0.831, $R^2$ = .008, $r$ = .219). However, the performance of the three predictor sets in unregularized models was extremely close to the criterion. The 2 items that were selected in 90% and 80% of the folds performed slightly better (MSE = 0.792, $R^2$ = .049, $r$ = .299) than the 5 items that were selected in 70% of the folds (MSE = 0.793, $R^2$ = .056, $r$ = .292). Consequently, 2 items were selected as most important predictors for the intensity of initiation conflicts.

## Intensity persistence conflicts

The aggregated performance across 100 folds of the regularized models using all trait predictors was below the minimal performance criterion (MSE = 1.090, $R^2$ = .006, $r$ = .206). However, the performance of the three predictor sets in unregularized models was satisfactory. The 18 items that were selected in 80% of the folds performed better (MSE = 1.001, $R^2$ = .084, $r$ = .360) than the 9 items that were selected in 90% folds (MSE = 1.001, $R^2$ = .075, $r$ = .336) and the 22 items that were selected in 70% of the folds (MSE = 1.021, $R^2$ = .061, $r$ = .335). Consequently, 18 items were selected as most important predictors of the intensity of persistence conflicts.

## Intensity inhibition conflicts

The aggregated performance across 100 folds of the regularized models using all trait predictors was very poor (MSE = 1.427, $R^2$ = -.041, $r$ = .028). No trait predictor was selected in at least 70% of the folds and thus the analysis had to stop here for the intensity of inhibition conflicts.

## Success of initiation conflicts

The aggregated performance across 100 folds of the regularized models using all trait predictors was satisfactory (MSE = 1.415, $R^2$ = .086, $r$ = .347). The 22 items that were selected in 90% of the folds performed better (MSE = 1.355, $R^2$ = .126, $r$ = .405) than the 27 items that were selected in 80% of the folds (MSE = 1.380, $R^2$ = .105, $r$ = .393) and the 32 items that were selected in 70% of the folds (MSE = 1.405, $R^2$ = .089, $r$ = .379). Consequently, 22 items were selected as most important predictors for the success of initiation conflicts.

### Success of persistence conflicts

The aggregated performance across 100 folds of the regularized models using all trait predictors was slightly below the minimal performance criterion (MSE = 1.484, $R^2$ = .036, $r$ = .283). However, the performance of the unregularized models using three predictor sets was satisfactory. The 14 items that were selected in 80% of the folds performed very slightly better (MSE = 1.418, $R^2$ = .082, $r$ = .358) than the 13 items that were selected in 90% of the folds (MSE = 1.407, $R^2$ = .076, $r$ = .358) and the 17 items that were selected in 70% of the folds (MSE = 1.439, $R^2$ = .066, $r$ = .341). Consequently, 14 items were selected as most important predictors for the success in persistence conflicts.

### Success of inhibition conflicts

The aggregated performance across 100 folds of the regularized models using all trait predictors was very poor (MSE = 1.979, $R^2$ = -.025, $r$ = .048). Since again no predictor was selected in at least 70% of the folds, we could not continue with the analysis for the prediction of the success of inhibition conflicts.

## Dimension reduction using EFA

### Frequency across self-control conflicts

The 13 selected items identified as the best predictors of the frequency across self-control conflicts met the requirements for the EFA as indicated by the Bartlett test ($\chi^2(78)$ = 538.88, $p < .001$). Results from parallel analysis suggested retaining three factors, whereas the MAP test recommended a single-factor solution. Since our goal was to examine nuanced relations between trait items and the frequency of self-control conflicts, we chose to disregard the MAP test recommendation and explored a three-factor solution, which we discuss below. However, we begin with the presentation of the two-factor solution, which we ultimately selected as the basis for computing composite scores.

Table S2 presents the results of the two-factor EFA. Factor 1 was labeled ‘dutifulness and compliance’ as it consisted of four items reflecting a tendency to care for personal belongings, be reliable, follow rules, and regulate one’s behavior in line with social expectations. Factor 2 was labeled ‘impulse inhibition’ as it compromised two items reflecting successful inhibition of undesired impulses. Moreover, six items were kept separate in further analysis as they did not fit into any of these factors. This concerns an item each capturing inattention (bisf14), reappraisal (ERQ5), successful behavior initiaiton (hoyle12), metacognitive regulation (MISCS25) and metacognitive knowlegde (MISCS4).

Table S3 shows the three-factor EFA solution as suggested by parallel analysis. Compared to the two-factor solution, both factors remained unchanged. Thus, the biggest difference was that the talkativeness item (bfi1) formed an own factor. Yet, given that this third factor was only compromised of a single item with a strong main loading, we concluded that the data does not support a three-factor structure.

**Table S2**

*EFA results with two factors of the 13 items identified as the best predictors of the frequency across self-control conflicts* ***(final solution)***

| Scale (Subscale) | Item name | Item text | Factor loadings | | $r_{outcome}$ |
|---|---|---|---|---|---|
| | | | **Dutifulness & Compliance** | **Impulse inhibition** | |
| BFI-K (extraversion) | bfi1 | Tends to be quiet. (R)* | | -0.14 | 0.10 |
| BIMS (attentional impulsivity) | bisf14 | I don't pay attention.* | -0.34 | -0.18 | -0.04 |
| ERQ (reappraisal) | ERQ5 | When I'm faced with a stressful situation, I make myself think about it in a way that helps me stay calm.* | -0.19 | 0.37 | 0.07 |
| CSCS (initiation) | hoyle12 | Even when the list of things to do is long, it is easy for me to get started.* | 0.24 | 0.24 | -0.19 |
| CSCS (inhibition) | hoyle3 | **I can deny myself something I want but don't need.** | | **0.45** | -0.18 |
| CSCS (inhibition) | hoyle6 | **I am able to control how I react to impulses.** | | **0.62** | -0.17 |
| IPIP (orderliness) | ipc2_2 | Like to tidy up.* | 0.33 | | 0.09 |
| IPIP (orderliness) | ipc2_5 | **don´t take good care of my belongings. (R)** | **0.43** | | -0.19 |
| IPIP (dutifulness) | ipc3_5 | **Keep my promises.** | **0.41** | | -0.15 |
| IPIP (dutifulness) | ipc3_6 | **Do the opposite of what is asked. (R)** | **0.63** | | -0.20 |
| MISCS (metacognitive regulation) | MISCS25 | I ask myself how well I accomplished my goals once I've resolved a self-control conflict.* | | 0.27 | 0.07 |
| MISCS (metacognitive knowledge) | MISCS4 | I understand my strengths and weaknesses when dealing with self-control conflicts.* | 0.14 | 0.21 | -0.17 |
| SCS | scs04 | **I say inappropriate things. (R)** | **0.37** | 0.23 | -0.17 |

*Note.* BFI-K: short version of the Big Five Inventory; BIMS: The Barrat Impulsiveness Scale; ERQ: Emotion Regulation Questionnaire; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

**Table S3**

*EFA results with three factors of the 13 items identified as the best predictors of the frequency across self-control conflicts*

| Scale (Subscale) | Item name | Item text | Factor loadings | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|
| | | | **Dutifulness & Compliance** | **Impulse inhibition** | **Talkative-ness** | |
| BFI-K (extraversion) | bfi1 | **Tends to be quiet. (R)** | | | **0.40** | 0.10 |
| BIMS (attentional impulsivity) | bisf14 | I don't pay attention.* | -0.33 | -0.17 | | -0.04 |
| ERQ (reappraisal) | ERQ5 | When I'm faced with a stressful situation, I make myself think about it in a way that helps me stay calm.* | -0.20 | 0.37 | | 0.07 |
| CSCS (initiation) | hoyle12 | Even when the list of things to do is long, it is easy for me to get started.* | 0.24 | 0.26 | 0.10 | -0.19 |
| CSCS (inhibition) | hoyle3 | **I can deny myself something I want but don't need.** | | **0.46** | | -0.18 |
| CSCS (inhibition) | hoyle6 | **I am able to control how I react to impulses.** | | **0.61** | | -0.17 |
| IPIP (orderliness) | ipc2_2 | Like to tidy up.* | 0.33 | | | 0.09 |
| IPIP (orderliness) | ipc2_5 | **don´t take good care of my belongings. I** | **0.43** | | | -0.19 |
| IPIP (dutifulness) | ipc3_5 | **Keep my promises.** | **0.42** | | | -0.15 |
| IPIP (dutifulness) | ipc3_6 | **Do the opposite of what is asked. I** | **0.63** | | | -0.20 |
| MISCS (metacognitive regulation) | MISCS25 | I ask myself how well I accomplished my goals once I've resolved a self-control conflict.* | | 0.29 | | 0.07 |
| MISCS (metacognitive knowledge) | MISCS4 | I understand my strengths and weaknesses when dealing with self-control conflicts.* | 0.16 | 0.29 | 0.35 | -0.17 |
| SCS | scs04 | I say inappropriate things. (R)* | 0.38 | 0.18 | -0.31 | -0.17 |

*Note.* BFI-K: short version of the Big Five Inventory; BIMS: The Barrat Impulsiveness Scale; ERQ: Emotion Regulation Questionnaire; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

## Intensity across self-control conflicts

The 20 selected items met the requirements for the EFA as indicated by the Bartlett test ($\chi^2(190) = 3047.59$ $p < .001$). Results from parallel analysis suggested retaining five factors, whereas the MAP test suggested retaining two factors. We will discuss both factor-solutions below, but begin with four factor-solution that we chose as final solution.

Table S3 displays the results of the EFA with four factors. As for the frequency across self-control conflicts, there were factors that captured successful self-control in a classical sense: Factor 1 comprised six items that capture successful 'behavior initiation', and Factor 3 comprised two items capturing 'resisting temptations'. In contrast to the frequency across self-control conflicts, we found a 'worry' factor that was composed of five items reflecting the tendency to worry and sensitivity to criticism. Moreover, as for the frequency across self-control conflicts, among the final items were items capturing metacognitive knowledge and regulation, which again had correlations with the outcome in the opposite direction. Thus, only the two metacognitive regulation items were merged into a 'metacognitive regulation' factor, whereas the metacognitive knowledge item was kept separate. Lastly, there were four items capturing talkativeness (bfi1), perfectionism (ipc2_4), self-confidence (ipc1_5), and the tendency to be driven (bisbas3) that did not fit into any of the four factors.

The results of the EFA with five factors are displayed in Table S4. The fifth factor does not add a new theme but instead seems to be a combination of the behavior initiation and worry factor, as for each of these two factors, multiple items show substantial cross-loadings onto the fifth factor. Thus, this factor solution was disregarded.

The results of the EFA with two factors are displayed in Table S5. In this solution, the items from the behavior initiation and resisting temptation factors formed a factor capturing 'capacity for self-control'. In addition, there was again a 'worry factor'. Thus, the greatest difference to the five-factor solution is the absence of the metacognitive-regulation factor. Based on the three-factor solution, both meta-regulation items would be kept separate in further analysis. Since these items capture very similar content, keeping them distinct did not appear to as the most parsimonious solution.

**Table S3**
*EFA results with four factors of the 20 items identified as the best predictors of the intensity across self-control conflicts* ***(final solution)***

| Scale (Subscale) | Item name | Item text | Factor loadings | | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|---|
| | | | **Behavior initiation** | **Worry** | **Resisting temptations** | **Meta-cognitive regulation** | |
| BFI-K (extraversion) | bfi1 | Tends to be quiet.* | | -0.11 | -0.13 | | -0.11 |
| BFI-K (neuroticism) | bfi10 | **Worries a lot.** | | **0.56** | | | 0.21 |
| BIS | bisbas13 | **I feel pretty worried or upset when I think or know somebody is angry at me.** | | **0.62** | | | 0.22 |
| BIS | bisbas19 | **I feel worried when I think I have done poorly at something.** | | **0.52** | 0.11 | | 0.17 |
| BIS | bisbas24 | **I worry about making mistakes.** | | **0.76** | | | 0.27 |
| BAS (drive) | bisbas3 | I work hard to reach my goals.* | 0.49 | 0.16 | 0.15 | 0.17 | 0.06 |
| BIS | bisbas8 | **Criticism or scolding hurts me quite a bit.** | | **0.70** | | | 0.22 |
| CSCS (initiation) | hoyle12 | **Even when the list of things to do is long, it is easy for me to get started.** | **0.64** | | | | -0.20 |
| CSCS (inhibition) | hoyle3 | **I can deny myself something I want but don't need. (R)** | | | **0.77** | | -0.15 |
| CSCS (inhibition) | hoyle4 | **My bad habits cause problems for me. (R)** | **0.45** | -0.15 | 0.17 | | -0.21 |
| IPIP (competence) | ipc1_5 | Am sure of my ground.* | 0.13 | -0.27 | 0.19 | 0.1 | -0.17 |
| IPIP (orderliness) | ipc2_4 | Like everything to be perfect.* | 0.26 | 0.34 | | | 0.17 |
| IPIP (self-discipline) | ipc5_6 | **Need a push to get started. (R)** | **0.74** | | | | -0.17 |
| IPIP (self-discipline) | ipc5_7 | **Have difficulty starting tasks. (R)** | **0.84** | | | | -0.19 |
| IPS | ips1 | **I put things off so long that my well-being or efficiency unnecessarily suffers.** | **-0.88** | | | | 0.18 |
| MSICS (metacognitive knowledge) | MISCS21 | I am a good judge of how well I resolve self-control conflicts.* | 0.13 | | | 0.24 | -0.17 |
| MSICS (metacognitive regulation) | MISCS23 | **I find myself pausing regularly to check how well I am resolving a self-control conflict.** | | | | **0.59** | 0.06 |
| MSICS (metacognitive regulation) | MISCS25 | **I ask myself how well I accomplished my goals once I've resolved a self-control conflict.** | | | | **0.78** | 0.08 |
| SCS | scs01 | **I am good at resisting temptation.** | 0.16 | | **0.70** | | -0.20 |
| SCS | scs06 | **I wish I had more self-discipline. (R)** | **0.60** | | 0.15 | | -0.18 |

*Note.* BFI-K: short version of the Big Five Inventory; BIS: The Behavioral Inhibition Scale; BAS: The Behavioral Activation Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

**Table S4**
*EFA results with five factors of the 20 items identified as the best predictors of the intensity across self-control conflicts*

| Scale (Subscale) | Item name | Item text | Factor loadings | | | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|---|---|
| | | | **Behavior initiation** | **Worry** | **Resisting temp-tations** | **Meta-cognitive regulation** | **?** | |
| BFI-K (extraversion) | bfi1 | Tends to be quiet.* | | -0.13 | -0.15 | | | -0.11 |
| BFI-K (neuroticism) | bfi10 | **Worries a lot.** | | **0.62** | | | -0.17 | 0.21 |
| BIS | bisbas13 | **I feel pretty worried or upset when I think or know somebody is angry at me.** | | **0.57** | | | 0.18 | 0.22 |
| BIS | bisbas19 | **I feel worried when I think I have done poorly at something.** | | **0.46** | | | 0.32 | 0.17 |
| BIS | bisbas24 | **I worry about making mistakes.** | | **0.79** | | | | 0.27 |
| BAS (drive) | bisbas3 | I work hard to reach my goals.* | 0.30 | | | 0.13 | 0.51 | 0.06 |
| BIS | bisbas8 | **Criticism or scolding hurts me quite a bit.** | | **0.66** | | | 0.12 | 0.22 |
| CSCS (initiation) | hoyle12 | **Even when the list of things to do is long, it is easy for me to get started.** | **0.64** | | | | | -0.20 |
| CSCS (inhibition) | hoyle3 | **I can deny myself something I want but don't need. (R)** | -0.10 | | **0.77** | | | -0.15 |
| CSCS (inhibition) | hoyle4 | **My bad habits cause problems for me. (R)** | **0.36** | -0.20 | 0.14 | | 0.21 | -0.21 |
| IPIP (competence) | ipc1_5 | Am sure of my ground.* | | -0.34 | 0.16 | | 0.26 | -0.17 |
| IPIP (orderliness) | ipc2_4 | Like everything to be perfect.* | 0.19 | 0.29 | | | 0.21 | 0.17 |
| IPIP (self-discipline) | ipc5_6 | **Need a push to get started. (R)** | **0.79** | | | | | -0.17 |
| IPIP (self-discipline) | ipc5_7 | **Have difficulty starting tasks. (R)** | **0.87** | | | | | -0.19 |
| IPS | ips1 | **I put things off so long that my well-being or efficiency unnecessarily suffers.** | **-0.82** | | | | -0.14 | 0.18 |
| MSICS (metacognitive knowledge) | MISCS21 | I am a good judge of how well I resolve self-control conflicts.* | | | | 0.24 | | -0.17 |
| MSICS (metacognitive regulation) | MISCS23 | **I find myself pausing regularly to check how well I am resolving a self-control conflict.** | | | | **0.70** | | 0.06 |
| MSICS (metacognitive regulation) | MISCS25 | **I ask myself how well I accomplished my goals once I've resolved a self-control conflict.** | -0.11 | | | **0.68** | | 0.08 |
| SCS | scs01 | **I am good at resisting temptation.** | 0.14 | | **0.73** | | | -0.20 |
| SCS | scs06 | **I wish I had more self-discipline. (R)** | **0.52** | -0.10 | 0.14 | | 0.21 | -0.18 |

*Note.* BFI-K: short version of the Big Five Inventory; BIS: The Behavioral Inhibition Scale; BAS: The Behavioral Activation Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

**Table S5**

*EFA results with two factors of the 20 items identified as the best predictors of the intensity across self-control conflicts*

| Scale (Subscale) | Item name | Item text | Factor loadings | | $r_{outcome}$ |
|---|---|---|---|---|---|
| | | | Capacity self-control | Worry | |
| BFI-K (extraversion) | bfi1 | Tends to be quiet.* | | | -0.11 |
| BFI-K (neuroticism) | bfi10 | **Worries a lot.** | | **0.56** | 0.21 |
| BIS | bisbas13 | **I feel pretty worried or upset when I think or know somebody is angry at me.** | | **0.62** | 0.22 |
| BIS | bisbas19 | **I feel worried when I think I have done poorly at something.** | | **0.51** | 0.17 |
| BIS | bisbas24 | **I worry about making mistakes.** | | **0.75** | 0.27 |
| BAS (drive) | bisbas3 | I work hard to reach my goals.* | 0.62 | 0.16 | 0.06 |
| BIS | bisbas8 | **Criticism or scolding hurts me quite a bit.** | | **0.70** | 0.22 |
| CSCS (initiation) | hoyle12 | **Even when the list of things to do is long, it is easy for me to get started.** | **0.67** | | -0.20 |
| CSCS (inhibition) | hoyle3 | **I can deny myself something I want but don't need. (R)** | **0.30** | | -0.15 |
| CSCS (inhibition) | hoyle4 | **My bad habits cause problems for me. (R)** | **0.54** | -0.16 | -0.21 |
| IPIP (competence) | ipc1_5 | Am sure of my ground.* | 0.27 | -0.28 | -0.17 |
| IPIP (orderliness) | ipc2_4 | Like everything to be perfect.* | 0.25 | 0.35 | 0.17 |
| IPIP (self-discipline) | ipc5_6 | **Need a push to get started. (R)** | **0.72** | | -0.17 |
| IPIP (self-discipline) | ipc5_7 | **Have difficulty starting tasks. (R)** | **0.79** | | -0.19 |
| IPS | ips1 | **I put things off so long that my well-being or efficiency unnecessarily suffers.** | **-0.78** | | 0.18 |
| MSICS (metacognitive knowledge) | MISCS21 | I am a good judge of how well I resolve self-control conflicts.* | 0.25 | | -0.17 |
| MSICS (metacognitive regulation) | MISCS23 | I find myself pausing regularly to check how well I am resolving a self-control conflict.* | 0.24 | | 0.06 |
| MSICS (metacognitive regulation) | MISCS25 | I ask myself how well I accomplished my goals once I've resolved a self-control conflict.* | 0.20 | | 0.08 |
| SCS | scs01 | **I am good at resisting temptation.** | **0.50** | -0.11 | -0.20 |
| SCS | scs06 | **I wish I had more self-discipline. (R)** | **0.69** | | -0.18 |

*Note.* BFI-K: short version of the Big Five Inventory; BIS: The Behavioral Inhibition Scale; BAS: The Behavioral Activation Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

## Success across self-control conflicts

The 17 selected items met the requirements for the EFA as indicated by the Bartlett test ($\chi^2(105) = 1425.91$, $p < .001$). Results from parallel analysis suggested retaining five factors, whereas the MAP test suggested retaining only a single factor. Again, we did not consisder a single factor solution due to the study's goals. The five factor solution as suggested by parallel analysis was tested and will be discussed below. However, we will begin with the presentation of a three-factor solution on which we ultimately based the assignment of items into composite scores.

The results of the EFA with three factors are displayed in Table S6. As for the other two components, there was a classical 'capacity for self-control' factor, compromised by eight items capturing achievement striving, perseverance, and the successful initiation of goal-directed behavior. In contrast to the other two components, no metacognitive regulation items were selected but only five metacognitive knowledge items, which formed a 'metacognitive knowledge' factor. In addition, there was an 'impulsive behavior' factor that consists of two items that measure the tendency to behave impulsively. Lastly, two items were kept separate because they did not fit into any of the factors: an item capturing honesty (ipic3_8) and an item capturing error avoidance (ipc6_2).

The results of the EFA with five factors are displayed in Table S7. In the five factor solution, we found three factors reflecting similar content as in the three-factor solution: capacity for self-control, metacognition, and impulsive behavior. However, three items that previoulsy loaded onto the capacity for self-control factor now formed two new factors. Specifically, two items formed a 'procrastination' factor, and one item formed a successful 'task completion' factor. Since each factor was only composed one or two items with strong main loadings and otherwise several items with relatively strong cross-loadings, we disregarded this factor solution.

**Table S6**

*EFA results with three factors of the 17 items identified as the best predictors of the success across self-control conflicts* ***(final solution)***

| Scale (subscale) | Item name | Item text | Factor loadings | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|
| | | | Capacity for self-control | Metacog-nition | Impulsive behavior | |
| BIMS (motor impulsivity) | bisf3 | **I do things without thinking.** | | | **0.90** | 0.04 |
| BIMS (motor impulsivity) | bisf5 | **I buy things on impulse.** | | | **0.36** | 0.08 |
| IPIP (dutifulness) | ipc3_8 | Misrepresent the facts. (R)* | 0.25 | | -0.24 | 0.17 |
| IPIP (achievement striving) | ipc4_3 | **Do just enough work to get by. (R)** | **0.54** | | | 0.24 |
| IPIP (achievement striving) | ipc4_6 | **Turn plans into actions.** | **0.58** | | | 0.32 |
| IPIP (self-discipline) | ipc5_2 | **Find it difficult to get down to work. (R)** | **0.68** | -0.1 | | 0.33 |
| IPIP (deliberation) | ipc6_2 | Avoid mistakes.* | 0.11 | | -0.26 | 0.15 |
| IPS | ips5 | **At the end of the day, I know I could have spent the time better. (R)** | **-0.58** | | | -0.28 |
| MISCS (metacognitive knowledge) | MISCS11 | **I am good at remembering past self-control conflicts** | | **0.51** | | 0.22 |
| MISCS (metacognitive knowledge) | MISCS12 | **I use different regulatory strategies depending on the situation.** | | **0.57** | | 0.19 |
| MISCS (metacognitive regulation) | MISCS15 | **I think of several strategies to resolve a self-control conflict and choose the best one.** | | **0.41** | | 0.19 |
| MISCS (metacognitive knowledge) | MISCS21 | **I am a good judge of how well I resolve self-control conflicts.** | | **0.46** | | 0.22 |
| MISCS (metacognitive knowledge) | MISCS4 | **I understand my strengths and weaknesses when dealing with self-control conflicts.** | | **0.46** | | 0.27 |
| SCS | scs09 | **I am able to work effectively toward long-term goals.** | **0.72** | | | 0.32 |
| STSC (start control) | stst10 | **I persevere at important tasks, even if I'm afraid something might go wrong.** | **0.58** | | | 0.29 |
| STSC (start control) | stst14 | **Even if I don't feel like it, I'm able to complete the tasks that needed to be done.** | **0.55** | | | 0.26 |
| STSC (start control) | stst16 | **When my mind wanders while I'm reading, it's easy for me to concentrate on the text again.** | **0.34** | 0.13 | | 0.25 |

*Note.* BIMS: The Barrat Impulsiveness Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale; STSC: The Stop and Start Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA.. Loadings < |.10| are not displayed.

**Table S7**

*EFA results with five factors of the 17 items identified as the best predictors of the success across self-control conflicts*

| Scale (subscale) | Item name | Item text | Factor loadings | | | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|---|---|
| | | | **Capacity for self-control** | **Meta-cognition** | **Task compl-etion** | **Pro-crasti-nation** | **Impulsive behavior** | |
| BIMS (motor impulsivity) | bisf3 | **I do things without thinking.** | | | | | **0.91** | 0.04 |
| BIMS (motor impulsivity) | bisf5 | **I buy things on impulse.** | 0.14 | | -0.12 | | **0.39** | 0.08 |
| IPIP (dutifulness) | ipc3_8 | Misrepresent the facts. (R)* | 0.32 | | | | -0.22 | 0.17 |
| IPIP (achievement striving) | ipc4_3 | **Do just enough work to get by. (R)** | **0.43** | | | 0.16 | | 0.24 |
| IPIP (achievement striving) | ipc4_6 | **Turn plans into actions.** | **0.56** | | | 0.10 | | 0.32 |
| IPIP (self-discipline) | ipc5_2 | **Find it difficult to get down to work. (R)** | 0.25 | | | **0.54** | | 0.33 |
| IPIP (deliberation) | ipc6_2 | Avoid mistakes.* | 0.21 | | | | -0.24 | 0.15 |
| IPS | ips5 | **At the end of the day, I know I could have spent the time better. (R)** | | | | **-0.69** | | -0.28 |
| MISCS (metacognitive knowledge) | MISCS11 | **I am good at remembering past self-control conflicts** | | **0.49** | | | | 0.22 |
| MISCS (metacognitive knowledge) | MISCS12 | **I use different regulatory strategies depending on the situation.** | | **0.53** | 0.13 | | | 0.19 |
| MISCS (metacognitive regulation) | MISCS15 | **I think of several strategies to resolve a self-control conflict and choose the best one.** | | **0.42** | | 0.13 | | 0.19 |
| MISCS (metacognitive knowledge) | MISCS21 | **I am a good judge of how well I resolve self-control conflicts.** | | **0.49** | | 0.16 | | 0.22 |
| MISCS (metacognitive knowledge) | MISCS4 | **I understand my strengths and weaknesses when dealing with self-control conflicts.** | 0.20 | **0.50** | -0.14 | | | 0.27 |
| SCS | scs09 | **I am able to work effectively toward long-term goals.** | **0.63** | | 0.12 | | | 0.32 |
| STSC (start control) | stst10 | I persevere at important tasks, even if I'm afraid something might go wrong.* | 0.46 | | 0.32 | | | 0.29 |
| STSC (start control) | stst14 | **Even if I don't feel like it, I'm able to complete the tasks that needed to be done.** | | | **0.93** | | | 0.26 |
| STSC (start control) | stst16 | When my mind wanders while I'm reading, it's easy for me to concentrate on the text again.* | | 0.17 | 0.20 | 0.33 | | 0.25 |

*Note.* BIMS: The Barrat Impulsiveness Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale; STSC: The Stop and Start Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

## Frequency of initiation conflicts

The 10 selected items met the requirements for the EFA as indicated by the Bartlett test ($\chi^2(45) = 532.39$, $p < .001$). Results from parallel analysis suggested retaining two factors, whereas the MAP test suggested retaining one factor.

The results of the two-factor solution are presented in Table S8. Factor 1 was termed 'capacity for self-control' as it contained not only the two items capturing successful impulse inhinition that were also selected for the frequency across self-control conflicts but additionally two items capturing successful behaviour initaition or preserverance. Notably, one item capturing metacognitive knowledge (MISCS4) also showed a substantial main loading onto this factor but was kept separate as it does not reflect self-control in a narrow sense as opposed to the other items of that factor. Factor 2 was termed 'dutifulness and compliance' and precisely resembeled the factor that that was also found for the frequency across self-control conflicts. Lasty, and again in alignment with the finding for the frequency across self-control conflicts, an item capturing inattention (bisf14) was kept separate as it did not fit into any of the factors.

**Table S8**

*EFA results with two factors of the 10 items identified as the best predictors of the frequency of initiation conflicts* ***(final solution)***

| Scale (Subscale) | Item name | Item text | Factor loadings: Capacity self-control | Factor loadings: Dutifulness & Compliance | $r_{outcome}$ |
|---|---|---|---|---|---|
| BIMS (attentional impulsivity) | bisf14 | I don't pay attention.* | -0.28 | -0.23 | -0.04 |
| CSCS (initiation) | hoyle12 | **Even when the list of things to do is long, it is easy for me to get started.** | **0.52** | | -0.17 |
| CSCS (inhibition) | hoyle3 | **I can deny myself something I want but don't need.** | **0.34** | 0.11 | -0.23 |
| IPIP (orderliness) | ipc2_5 | **don´t take good care of my belongings. I** | 0.12 | **0.33** | -0.19 |
| IPIP (dutifulness) | ipc3_1 | **Try to follow rules.** | | **0.52** | -0.18 |
| IPIP (deliberation) | ipc6_6 | **Rush into things. I** | | **0.51** | -0.19 |
| MSICS (metacognitive knowledge) | MISCS4 | I understand my strengths and weaknesses when dealing with self-control conflicts.* | 0.33 | | -0.16 |
| SCS | scs04 | **I say inappropriate things. (R)** | | **0.64** | -0.17 |
| STSC (start control) | stst10 | **I persevere at important tasks, even if I'm afraid something might go wrong.** | **0.49** | 0.10 | -0.17 |
| STSC (start control) | stst16 | **When my mind wanders while I'm reading, it's easy for my to concentrate on the text again.** | **0.60** | -0.12 | -0.14 |

*Note.* BIMS: The Barrat Impulsiveness Scale; IPIP: International Personality Item Pool; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale; STSC: The Stop and Start Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

## Frequency of persistence conflicts

The 11 selected items met the requirements for the EFA as indicated by the Bartlett test ($\chi^2(55) = 423.18$, $p < .001$). The results of the parallel analysis suggested retaining four factors, whereas the MAP test suggested retaining one factor. The four-factor solution as suggested by parallel analysis was tested and will be discussed below. However, we will begin with the presentation of a two-factor solution on which we ultimately based the assignment of items into composite scores.

The results of the EFA with two factors are displayed in Table S9. Factor 1 was termed 'regulation' as it consisted of five items capturing emotional reappraisal, metacognitive regulation, and metacognitive knowledge with a regulatory component. In line with the results for the frequency of self-control conflicts and initiation conflicts, Factor 2 was labeled 'diligence and compliance' even though this time, it was only composed of two items capturing the tendency to care for personal belongings and follow rules. Four items were kept separate in further analysis due to substantial loadings on multiple factors, a bivariate correlation with the outcome diverged from other items in the same factor or because they did not load substantially onto any of the factors. This concerns an item capturing the tendency to be tidy (ipc2_2), the tendency to postpone important tasks (ips2), talkativeness (bfi1), and a metacognitive knowledge item (MISCS4).

The results of the EFA with four factors are displayed in Table S10. In this solution, the regulation factor from the two-factor solution is split into two separate factors for emotional reappraisal and metacognitive regulation. In contrast, the dutifulness and compliance factor remains unchanged compared to the two-factor solution. Several items demonstrate substantial cross-loadings onto the fourth factor, with only a metacognitive knowledge item representing a substantial main-loading onto this factor. Thus, the fourth-factor was not well-defined. When exploring a three-factor solution (Table S11), we found the same three factors that were realtively well-defined in the four-factor solution. However, one of the two items of the reappraisal factor loaded almost equally strong onto the metacognitive regulation factor, suggesting that these factors may not be sufficiently distinct. Thus, we tested the above discussed two-factor solution and conclude that there are only two well-defined factors amont the final items for the frequency of persistence conflicts.

**Table S9**

*EFA results with two factors of the 11 items identified as the best predictors of the frequency of persistence conflicts* ***(final solution)***

| Scale (subscale) | Item name | Item text | Factor loadings: Regulation | Factor loadings: Diligence & compliance | $r_{outcome}$ |
|---|---|---|---|---|---|
| BFI-K (extraversion) | bfi1 | Tends to be quiet. (R)* | | | 0.11 |
| ERQ (reappraisal) | ERQ3 | **When I want to feel less negative emotion (such as sadness or anger), I change what I'm thinking about** | **0.32** | | 0.11 |
| ERQ (reappraisal) | ERQ5 | **When I'm faced with a stressful situation, I make myself think about it in a way that helps me stay calm.** | **0.39** | -0.13 | 0.18 |
| IPIP (orderliness) | ipc2_2 | Like to tidy up.* | | 0.40 | 0.10 |
| IPIP (orderliness) | ipc2_5 | **don´t take good care of my belongings. ®** | | **0.55** | -0.12 |
| IPIP (dutifulness) | ipc3_6 | **Do the opposite of what is asked.** | | **0.45** | -0.14 |
| IPS | ips2 | If there is something I should do, I get to it before attending to lesser tasks. (R)* | -0.17 | -0.21 | -0.09 |
| MISCS (metacognitive knowledge) | MISCS12 | **I use different regulatory strategies depending on the situation.** | **0.50** | | 0.12 |
| MISCS (metacognitive regulation) | MISCS19 | **I find myself analyzing the usefulness of strategies during self-control conflicts.** | **0.55** | | 0.14 |
| MISCS (metacognitive regulation) | MISCS23 | **I find myself pausing regularly to check how well I am resolving a self-control conflict.** | **0.53** | | 0.14 |
| MISCS (metacognitive knowledge) | MISCS4 | I understand my strengths and weaknesses when dealing with self-control conflicts.* | 0.35 | 0.20 | -0.09 |

*Note.* BFI-K: short version of the Big Five Inventory; ERQ: Emotion Regulation Questionnaire; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

**Table S10**

*EFA results with four factors of the 11 items identified as the best predictors of the frequency of persistence conflicts*

| Scale (subscale) | Item name | Item text | Factor loadings | | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|---|
| | | | **Reappraisal** | **Metacognitive regulation** | **Dutifulness & compliance** | **Metacognitive knowledge** | |
| BFI-K (extraversion) | bfi1 | tends to be quiet. (R)* | | -0.14 | -0.17 | 0.29 | 0.11 |
| ERQ (reappraisal) | ERQ3 | **When I want to feel less negative emotion (such as sadness or anger), I change what I'm thinking about** | **0.31** | | | 0.24 | 0.11 |
| ERQ (reappraisal) | ERQ5 | **When I'm faced with a stressful situation, I make myself think about it in a way that helps me stay calm.** | **1.00** | | | | 0.18 |
| IPIP (orderliness) | ipc2_2 | Like to tidy up.* | | 0.11 | 0.40 | | 0.10 |
| IPIP (orderliness) | ipc2_5 | **don´t take good care of my belongings. ®** | | | **0.61** | | -0.12 |
| IPIP (dutifulness) | ipc3_6 | **Do the opposite of what is asked.** | | | **0.39** | 0.11 | -0.14 |
| IPS | ips2 | If there is something I should do, I get to it before attending to lesser tasks. (R)* | | | -0.14 | -0.26 | -0.09 |
| MISCS (metacognitive knowledge) | MISCS12 | **I use different regulatory strategies depending on the situation.** | | **0.22** | | 0.35 | 0.12 |
| MISCS (metacognitive regulation) | MISCS19 | **I find myself analyzing the usefulness of strategies during self-control conflicts.** | | **0.68** | | | 0.14 |
| MISCS (metacognitive regulation) | MISCS23 | **I find myself pausing regularly to check how well I am resolving a self-control conflict.** | | **0.59** | | | 0.14 |
| MISCS (metacognitive knowledge) | MISCS4 | I understand my strengths and weaknesses when dealing with self-control conflicts.* | | | | **0.54** | -0.09 |

*Note.* BFI-K: short version of the Big Five Inventory; ERQ: Emotion Regulation Questionnaire; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

**Table S11**

*EFA results with three factors of the 11 items identified as the best predictors of the frequency of persistence conflicts*

| Scale (subscale) | Item name | Item text | Factor loadings | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|
| | | | Metacognitive Regulation | Reappraisal | Dutifulness & compliance | |
| BFI-K (extraversion) | bfi1 | tends to be quiet. (R)* | | | | 0.11 |
| ERQ (reappraisal) | ERQ3 | **When I want to feel less negative emotion (such as sadness or anger), I change what I'm thinking about** | | **1.00** | | 0.11 |
| ERQ (reappraisal) | ERQ5 | **When I'm faced with a stressful situation, I make myself think about it in a way that helps me stay calm.** | 0.24 | **0.28** | | 0.18 |
| IPIP (orderliness) | ipc2_2 | Like to tidy up.* | | | 0.40 | 0.10 |
| IPIP (orderliness) | ipc2_5 | **don´t take good care of my belongings. (R)** | | | **0.55** | -0.12 |
| IPIP (dutifulness) | ipc3_6 | **Do the opposite of what is asked.** | | | **0.45** | -0.14 |
| IPS | ips2 | If there is something I should do, I get to it before attending to lesser tasks. (R)* | -0.15 | | -0.22 | -0.09 |
| MISCS (metacognitive knowledge) | MISCS12 | **I use different regulatory strategies depending on the situation.** | **0.46** | | | 0.12 |
| MISCS (metacognitive regulation) | MISCS19 | **I find myself analyzing the usefulness of strategies during self-control conflicts.** | **0.64** | | | 0.14 |
| MISCS (metacognitive regulation) | MISCS23 | **I find myself pausing regularly to check how well I am resolving a self-control conflict.** | **0.58** | | | 0.14 |
| MISCS (metacognitive knowledge) | MISCS4 | I understand my strengths and weaknesses when dealing with self-control conflicts.* | 0.28 | 0.12 | 0.23 | -0.09 |

*Note.* BFI-K: short version of the Big Five Inventory; ERQ: Emotion Regulation Questionnaire; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

## Intensity of initiation conflicts

Since only two items were selected, conducing an EFA to reduce the dimensionality was deemed not necessary. The two items capture successful behavior initiation ("I have difficulty starting tasks", recoded) and the tendency to worry ("I worry about making mistakes").

## Intensity of persistence conflicts

The 18 selected items met the requirements for the EFA as indicated by the Bartlett test ($\chi^2(153) = 1649.62$, $p < .001$). Parallel analysis suggested retaining five factors, whereas the MAP test suggested retaining two factors. We present and discuss both the five- and two-factor solution below. However, we begin with the four-factor solution which we ultimately chose to compute the composite scores.

The results of the EFA with four factors are depicted in Table S11. The first three factors closely resemble those found for the intensity across self-control conflicts. Factor 1 was termed 'worry' and consisted of five items capturing the tendency to worry, to be afraid or the sensitivity to criticism. Reflecting self-control in a classical sense, Factor 2 was composed of three items capturing successful 'behavior initiation', and Factor 3 was composed of two items capturing successful 'temptation resistance'. Three additional items displayed a substantial main loading onto the behavior intiation factor but were kept separate in further analyses due to their correlation with the outcome being in the opposite direction compared to the other items. These included an item capturing determined drive (bisbas9) and an item capturing the tendency to postpone important tasks (ips2). In contrast to the results for the intensity across self-control conflicts, there was a factor capturing 'social impulsivity,' composed of three items reflecting talkativeness, a tendency to say inappropriate things, and a lack of adherence to social expectations. Lastly, there were three items that either lay directly between both factors or did not fit clearly into either one. This applies to items capturing self-confidence (ipc1_5), perfectionism (ipc2_4), and metacognitive knowlegde (MISCS21).

The results of the EFA with five factors are displayed in Table S12. Here, the same factors as in the four-factor solution were found. Yet, the fifth factor did not have any items with clear main loadings, which is why this factor solution was disregarded.

The results of the EFA with two factors are displayed in Table S13. Here, the items of the behavior intiation and temptation resistance factor formed together a factor caturing the 'capacity for self-control'. In addition, the 'worry' factor was found again. In this solution, all three items from the social impulsivity factor of the four-factor solution would have to be kept separate. Since we aimed for parsimonity, we preferred the four-factor solution.

**Table S11**

*EFA results with four factors of the 18 items identified as the best predictors of the intensity of persistence conflicts* ***(final solution)***

| Scale (subscale) | Item name | Item text | Factor loadings | | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|---|
| | | | Worry | Behavior initiation | Temptation resistance | Social impulsivity | |
| BFI-K (extraversion) | bfi1 | **tends to be quiet. (R)** | | 0.13 | | **-0.36** | -0.15 |
| BIS | bisbas13 | **I feel pretty worried or upset when I think or know somebody is angry at me.** | **0.63** | | | | 0.18 |
| BIS | bisbas19 | **I feel worried when I think I have done poorly at something.** | **0.57** | 0.14 | 0.13 | -0.11 | 0.20 |
| BIS | bisbas22 | **I have very few fears compared to my friends.** | **0.53** | | | 0.21 | 0.18 |
| BIS | bisbas24 | **I worry about making mistakes.** | **0.71** | | | 0.11 | 0.22 |
| BIS | bisbas8 | **Criticism or scolding hurts me quite a bit.** | **0.73** | | | | 0.22 |
| BAS (drive) | bisbas9 | When I want something, I usually go all-out to get it.* | | 0.53 | -0.13 | -0.20 | 0.09 |
| CSCS (inhibition) | hoyle1 | **I am able to resist temptations** | | 0.24 | **0.59** | | -0.13 |
| CSCS (inhibition) | hoyle3 | **I can deny myself something I want but don't need.** | | | **0.81** | | -0.11 |
| CSCS (inhibition) | hoyle4 | **My bad habits cause problems for me.** | -0.21 | **0.38** | 0.13 | 0.20 | -0.14 |
| IPIP (competence) | ipc1_5 | Am sure of my ground.* | -0.27 | 0.29 | 0.14 | | -0.19 |
| IPIP (orderliness) | ipc2_4 | Like everything to be perfect.* | 0.27 | 0.31 | -0.13 | | 0.13 |
| IPIP (self-discipline) | ipc5_6 | **Need a push to get started. I** | | **0.44** | 0.10 | 0.22 | -0.12 |
| IPS | ips2 | If there is something I should do, I get to it before attending to lesser tasks.* | | -0.49 | -0.12 | | -0.10 |
| IPS | ips9 | **I do everything when I believe it needs to be done. (R)** | | **-0.62** | | | 0.11 |
| MISCS (metacognitive knowlegde) | MISCS21 | I am a good judge of how well I resolve self-control conflicts.* | | 0.21 | 0.10 | | -0.16 |
| SCS | scs04 | **I say inappropriate things. (R)** | 0.10 | 0.14 | 0.10 | **0.40** | 0.11 |
| STST | stst4 | **I stick to the rules even if I find them unreasonable.** | 0.12 | 0.13 | | **0.36** | 0.11 |

*Note.* BFI-K: short version of the Big Five Inventory; BIS: The Behavioral Inhibition Scale; BAS: The Behavioral Activation Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale; STSC: The Stop and Start Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

**Table S12**
*EFA results with five factors of the 18 items identified as the best predictors of the intensity of persistence conflicts*

| Scale (subscale) | Item name | Item text | Factor loadings | | | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|---|---|
| | | | Worry | Behavior initiation | Temptation resistance | ? | Social impulsivity | |
| BFI-K (extraversion) | bfi1 | **tends to be quiet. (R)** | | 0.15 | | | **-0.48** | -0.15 |
| BIS | bisbas13 | **I feel pretty worried or upset when I think or know somebody is angry at me.** | **0.65** | | | | | 0.18 |
| BIS | bisbas19 | **I feel worried when I think I have done poorly at something.** | **0.59** | | 0.14 | 0.10 | | 0.20 |
| BIS | bisbas22 | **I have very few fears compared to my friends.** | **0.47** | 0.16 | | -0.31 | | 0.18 |
| BIS | bisbas24 | **I worry about making mistakes.** | **0.68** | | | -0.13 | | 0.22 |
| BIS | bisbas8 | **Criticism or scolding hurts me quite a bit.** | **0.72** | | | | | 0.22 |
| BAS (drive) | bisbas9 | **When I want something, I usually go all-out to get it.** | 0.12 | 0.27 | -0.14 | 0.45 | -0.10 | 0.09 |
| CSCS (inhibition) | hoyle1 | **I am able to resist temptations** | | 0.28 | **0.60** | | | -0.13 |
| CSCS (inhibition) | hoyle3 | **I can deny myself something I want but don't need.** | | | **0.78** | | | -0.11 |
| CSCS (inhibition) | hoyle4 | **My bad habits cause problems for me.** | -0.21 | **0.41** | 0.12 | | 0.14 | -0.14 |
| IPIP (competence) | ipc1_5 | Am sure of my ground.* | -0.20 | | 0.13 | 0.49 | 0.13 | -0.19 |
| IPIP (orderliness) | ipc2_4 | Like everything to be perfect.* | 0.28 | 0.22 | -0.15 | 0.17 | 0.12 | 0.13 |
| IPIP (self-discipline) | ipc5_6 | **Need a push to get started. I** | | **0.57** | | | | -0.12 |
| IPS | ips2 | If there is something I should do, I get to it before attending to lesser tasks.* | | -0.51 | -0.12 | | 0.12 | -0.10 |
| IPS | ips9 | **I do everything when I believe it needs to be done. (R)** | | **-0.58** | | -0.14 | | 0.11 |
| MISCS (metacognitive knowlegde) | MISCS21 | I am a good judge of how well I resolve self-control conflicts.* | | | | 0.20 | | -0.16 |
| SCS | scs04 | **I say inappropriate things. (R)** | | 0.13 | | | **0.46** | 0.11 |
| STST | stst4 | **I stick to the rules even if I find them unreasonable.** | | 0.17 | | | **0.34** | 0.11 |

*Note.* BFI-K: short version of the Big Five Inventory; BIS: The Behavioral Inhibition Scale; BAS: The Behavioral Activation Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale; STSC: The Stop and Start Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

**Table S13**

*EFA results with two factors of the 18 items identified as the best predictors of the intensity of persistence conflicts*

| Scale (subscale) | Item name | Item text | Factor loadings | | $r_{outcome}$ |
|---|---|---|---|---|---|
| | | | Worry | Capacity for self-control | |
| BFI-K (extraversion) | bfi1 | tends to be quiet. (R)* | -0.13 | | -0.15 |
| BIS | bisbas13 | **I feel pretty worried or upset when I think or know somebody is angry at me.** | **0.61** | | 0.18 |
| BIS | bisbas19 | **I feel worried when I think I have done poorly at something.** | **0.52** | 0.14 | 0.20 |
| BIS | bisbas22 | **I have very few fears compared to my friends.** | **0.58** | | 0.18 |
| BIS | bisbas24 | **I worry about making mistakes.** | **0.73** | | 0.22 |
| BIS | bisbas8 | **Criticism or scolding hurts me quite a bit.** | **0.70** | | 0.22 |
| BAS (drive) | bisbas9 | When I want something, I usually go all-out to get it.* | | 0.28 | 0.09 |
| CSCS (inhibition) | hoyle1 | **I am able to resist temptations** | -0.11 | **0.59** | -0.13 |
| CSCS (inhibition) | hoyle3 | **I can deny myself something I want but don't need.** | | **0.36** | -0.11 |
| CSCS (inhibition) | hoyle4 | **My bad habits cause problems for me.** | -0.13 | **0.54** | -0.14 |
| IPIP (competence) | ipc1_5 | Am sure of my ground.* | -0.27 | 0.34 | -0.19 |
| IPIP (orderliness) | ipc2_4 | Like everything to be perfect.* | 0.32 | 0.22 | 0.13 |
| IPIP (self-discipline) | ipc5_6 | **Need a push to get started. I** | | **0.57** | -0.12 |
| IPS | ips2 | If there is something I should do, I get to it before attending to lesser tasks.* | | -0.50 | -0.10 |
| IPS | ips9 | **I do everything when I believe it needs to be done. (R)** | | **-0.58** | 0.11 |
| MISCS (metacognitive knowlegde) | MISCS21 | I am a good judge of how well I resolve self-control conflicts.* | | 0.23 | -0.16 |
| SCS | scs04 | I say inappropriate things. (R)* | 0.21 | 0.36 | 0.11 |
| STST | stst4 | I stick to the rules even if I find them unreasonable.* | 0.22 | 0.23 | 0.11 |

*Note.* BFI-K: short version of the Big Five Inventory; BIS: The Behavioral Inhibition Scale; BAS: The Behavioral Activation Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale; STSC: The Stop and Start Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

## Success of initiation conflicts

The 22 selected items met the requirements for the EFA as indicated by the Bartlett test ($\chi^2(231) = 3621.43$, $p < .001$). Results from parallel analysis suggested retaining seven factors, whereas the MAP test suggested retaining one factor. However, we begin with the presentation of the results of the EFA with five factors, which we chose as final solution.

Table S14 presents the results of the EFA with five factors. Factor 1 was labeled 'behavior initiation' as it comprised six items tapping into initiating goal-directed behavior and a lack of procrastination. Factor 2 consists of two items capturing concentration issues and was thus labeled 'attentional impulsivity'. Factor 3 comprised three items capturing 'impulsive behavior'. Factor 4 summarizes four items capturing 'metacognitive knowlegde'. Factor 5 was substantially less clear defined compared to the other factors as indicated by several substantial cross-loadings. However, the two items with the highest main loading both clearly captured the preserverance, which is why Factor 5 was composed of these two items and termed 'preserverance'. Lastly, there were five items that were kept separate due to substantial ladongs onto several factors. This concerns items capturing the following content: talkativeness (bf1), depression (bf11), persistence after success (bisbas4), plan implementation (ipc4_6), and successfully working towards long-term goals (scs09).

Table S15 presents the results of the EFA with seven factors as suggested by parallel analysis. Here, the first five factors were very similar to those found in the five-factor solution, while the additional two factors were not clearly defined.

**Table S14**

*EFA results with seven factors of the 22 items identified as the best predictors of the success of initiation conflicts **(final solution)***

| Scale (subscale) | Item name | Item text | Factor loadings | | | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|---|---|
| | | | **Behavior initiation** | **Attentional impulsivity** | **Impulsive behavior** | **Metacognitive knowledge** | **Preserverance** | |
| BFI-K (extraversion) | bfi1 | tends to be quiet. (R)* | | | 0.22 | 0.16 | | 0.13 |
| BFI-K (neuroticism) | bfi11 | tends to feel depressed, blue.* | -0.23 | -0.29 | | | 0.17 | -0.19 |
| BAS (reward responsiveness) | bisbas4 | When I'm doing well at something, I love to keep at it.* | | 0.23 | | | 0.20 | 0.20 |
| BIMS (attentional impulsivity) | bisf13 | **I concentrate easily. ®** | | **-0.79** | | | | -0.26 |
| BIMS (motor impulsivity) | bisf3 | **I do things without thinking.** | | | **0.41** | | -0.19 | 0.05 |
| BIMS (motor impulsivity) | bisf5 | **I buy things on impulse.** | | | **0.84** | | | 0.10 |
| CSCS (initiation) | hoyle13 | **I get started on new projects right away.** | **0.56** | | | | 0.14 | 0.26 |
| IPIP (achievement striving) | ipc4_6 | Turn plans into actions.* | 0.27 | 0.13 | | | 0.32 | 0.30 |
| IPIP (self-discipline) | ipc5_2 | **Find it difficult to get down to work. (R)** | **0.75** | 0.13 | | | | 0.30 |
| IPIP (self-discipline) | ipc5_5 | **Start tasks right away.** | **0.76** | -0.13 | | | 0.17 | 0.28 |
| IPIP (self-discipline) | ipc5_7 | **Have diffuculty starting tasks. (R)** | **0.85** | | | | | 0.30 |
| IPS | ips5 | **At the end of the day, I know I could have spent the time better.** | **-0.52** | -0.19 | | | | -0.24 |
| IPS | ips7 | **I delay tasks beyond what is reasonable.** | **-0.60** | | | | -0.21 | -0.24 |
| MISCS (metacognitive knowledge) | MISCS11 | **I am good at remembering past self-control conflicts** | | | | **0.50** | 0.11 | 0.18 |
| MISCS (metacognitive knowledge) | MISCS12 | **I use different regulatory strategies depending on the situation.** | | | | **0.52** | 0.10 | 0.18 |
| MISCS (metacognitive knowledge) | MISCS18 | **I am aware of what strategies I use to resolve self-control conflicts.** | | | | **0.67** | -0.12 | 0.19 |
| MISCS (metacognitive knowledge) | MISCS4 | **I understand my strengths and weaknesses when dealing with self-control conflicts.** | | | | **0.49** | | 0.21 |
| SCS | scs08 | **I have trouble concentrating. (R)** | | **0.83** | | | | 0.27 |
| SCS | scs09 | I am able to work effectively toward long-term goals.* | 0.34 | 0.11 | | 0.15 | 0.38 | 0.30 |
| STSC (stop-control) | stst1 | **During shopping I make impulsive purchases. (R)** | | | **-0.72** | | | -0.08 |
| STSC (start-control) | stst10 | **I persevere at important tasks, even if I'm afraid something might go wrong.** | | 0.20 | | | **0.61** | 0.26 |
| STSC (start-control) | stst14 | **Even if I don't feel like it, I'm able to complete the tasks that needed to be done.** | 0.10 | 0.25 | | | **0.43** | 0.25 |

*Note.* BFI-K: short version of the Big Five Inventory; BAS: The Behavioral Activation Scale; BIMS: The Barrat Impulsiveness Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale; STSC: The Stop and Start Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

**Table S15**
*EFA results with seven factors of the 22 items identified as the best predictors of the success of initiation conflicts*

| Scale (subscale) | Item name | Item text | Factor loadings | | | | | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | **Behavior initiation** | **Attentional impulsivity** | **Impulsive buying** | **Preserverance** | **Metacognitive knowledge** | **?** | **?** | |
| BFI-K (extraversion) | bfi1 | tends to be quiet. (R)* | | -0.19 | 0.12 | 0.14 | 0.13 | | 0.31 | 0.13 |
| BFI-K (neuroticism) | bfi11 | tends to feel depressed, blue.* | -0.13 | -0.18 | | 0.11 | | -0.27 | -0.36 | -0.19 |
| BAS (reward responsiveness) | bisbas4 | When I'm doing well at something, I love to keep at it.* | | 0.25 | 0.11 | | | 0.35 | -0.16 | 0.20 |
| BIMS (attentional impulsivity) | bisf13 | **I concentrate easily. ®** | | **-0.77** | | | | | | -0.26 |
| BIMS (motor impulsivity) | bisf3 | I do things without thinking.* | -0.11 | | 0.30 | | | | 0.45 | 0.05 |
| BIMS (motor impulsivity) | bisf5 | **I buy things on impulse.** | | | **0.79** | | | | | 0.10 |
| CSCS (initiation) | hoyle13 | **I get started on new projects right away.** | **0.55** | | | | | 0.11 | | 0.26 |
| IPIP (achievement striving) | ipc4_6 | **Turn plans into actions.** | 0.18 | | | 0.21 | | 0.46 | | 0.30 |
| IPIP (self-discipline) | ipc5_2 | **Find it difficult to get down to work. (R)** | **0.72** | 0.10 | | | | | 0.12 | 0.30 |
| IPIP (self-discipline) | ipc5_5 | **Start tasks right away.** | **0.78** | | | | | | -0.15 | 0.28 |
| IPIP (self-discipline) | ipc5_7 | **Have diffuculty starting tasks. (R)** | **0.84** | | | | | | | 0.30 |
| IPS | ips5 | **At the end of the day, I know I could have spent the time better.** | **-0.48** | -0.14 | | | | | -0.15 | -0.24 |
| IPS | ips7 | **I delay tasks beyond what is reasonable.** | **-0.62** | | | -0.14 | | | 0.15 | -0.24 |
| MISCS (metacognitive knowledge) | MISCS11 | **I am good at remembering past self-control conflicts** | | | 0.11 | | **0.49** | | | 0.18 |
| MISCS (metacognitive knowledge) | MISCS12 | **I use different regulatory strategies depending on the situation.** | | | | 0.18 | **0.51** | | | 0.18 |
| MISCS (metacognitive knowledge) | MISCS18 | **I am aware of what strategies I use to resolve self-control conflicts.** | | | | | **0.68** | | | 0.19 |
| MISCS (metacognitive knowledge) | MISCS4 | **I understand my strengths and weaknesses when dealing with self-control conflicts.** | | | | | **0.43** | 0.32 | | 0.21 |
| SCS | scs08 | **I have trouble concentrating. (R)** | | 0.88 | | | | | | 0.27 |
| SCS | scs09 | I am able to work effectively toward long-term goals.* | 0.26 | | | 0.37 | | 0.33 | | 0.30 |
| STSC (stop-control) | stst1 | **During shopping I make impulsive purchases. (R)** | | | **-0.79** | | | | | -0.08 |
| STSC (start-control) | stst10 | **I persevere at important tasks, even if I'm afraid something might go wrong.** | | | | **0.61** | | 0.16 | | 0.26 |
| STSC (start-control) | stst14 | **Even if I don't feel like it, I'm able to complete the tasks that needed to be done.** | | 0.16 | | **0.65** | | -0.11 | | 0.25 |

*Note.* BFI-K: short version of the Big Five Inventory; BAS: The Behavioral Activation Scale; BIMS: The Barrat Impulsiveness Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; IPS: The Irrational Procrastination Scale; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale; STSC: The Stop and Start Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

## Success of persistence conflicts

The 14 selected items met the requirements for the EFA as indicated by the Bartlett test ($\chi^2(91) = 1196{,}08$, $p < .001$). Results from parallel analysis suggested retaining four factors, whereas the MAP test suggested retaining one factor. However, we begin with the presentation of the results of the EFA with two factors, which we chose as final solution.

The results of the EFA with two factors are presented in Table S16. Factor 1 was labeled 'self-discipline' as it comprises five items that capture self-discipline, lack of procrastination and initiate goal-directed behavior despite difficulties. Factor 2 summarizes three items capturing meta-knowledge and regulation and was termed 'metacognition'. In addition, six items were kept separate as they did not clearly fit into any of these factors. This concerns items capturing the following content: depression (bf11), reward responsiveness (bisbas7), ease of persistence (hoyle20), competence (ipc1_4), refocusing after distraction (stst16), and persistence despite fatigue (stst17).

The results of the EFA with four factors are presented in Table S17. In this factor solution, there was again a self-discipline and a metacognition factor. In additon, four items capturing competence, reactivity to achievements, and depression, which may together reflect the 'positive outcome expectation' of an individual formed a third factor. However, the fourth factor was only composed of a single item with a clear main loading that relfects the ability to refocus after distraction (stst16). Since there seemed to be unsufficient support for the fourth factor, we additionally tested a solution with three factors that is presented in Table S18. Here, we again find a self-discipline and a metacognition factor. Factor 3 was very similar to the factor reflecting positive outcome expectations. Yet, in this factor solution, and item capturing competence additionally showed substantial loadings onto the self-discipline factor and would thus have to be kept separate, and the depression item also showed substantially cross-loadings. Thus, overall, we concluded that the factor was not clearly defined in the three-factor solution and thus preferred the two-factor solution.

**Table S16**

*EFA results with two factors of the 14 items identified as the best predictors of the success of persistence conflicts* ***(final solution)***

| Scale (subscale) | Name | Item text | Factor loadings | | $r_{outcome}$ |
|---|---|---|---|---|---|
| | | | **Self-disci-pline** | **Meta-cognition** | |
| BFI-K (neuroticism) | bfi11 | tends to feel depressed, blue.* | -0.41 | -0.16 | -0.23 |
| BAS (reward responsiveness) | bisbas7 | When I get something I want, I feel excited and energized.* | | 0.22 | 0.15 |
| CSCS (persistence) | hoyle20 | I find it easy to keep with good behavior.* | 0.32 | 0.26 | 0.20 |
| IPIP (competence) | ipc1_4 | Don't understand things. (R)* | 0.26 | 0.22 | 0.20 |
| IPIP (competence) | ipc1_8 | **Complete tasks successfully.** | **0.43** | 0.15 | 0.21 |
| IPIP (self-discipline) | ipc5_2 | **Find it difficult to get down to work.** | **0.72** | | 0.26 |
| IPS | ips5 | **At the end of the day, I know I could have spent the time better.** | **-0.69** | | -0.25 |
| MISCS (metacognitive knowledge) | MISCS12 | **I use different regulatory strategies depending on the situation.** | -0.12 | **0.48** | 0.20 |
| MISCS (metacognitive regulation) | MISCS15 | **I think of several strategies to resolve a self-control conflict and choose the best one.** | | **0.35** | 0.18 |
| MISCS (metacognitive knowlede) | MISCS4 | **I understand my strengths and weaknesses when dealing with self-control conflicts.** | | **0.54** | 0.22 |
| SCS | scs06 | **I wish I had more self-discipline. (R)** | **0.68** | | 0.23 |
| STSC (start control) | stst12 | **I find it difficult having to restart something after I thought I was already done. (R)** | **0.43** | | 0.20 |
| STSC (start control) | stst16 | When my mind wanders while I'm reading, It's easy for my to concentrate on the text again.* | 0.26 | 0.27 | 0.21 |
| STSC (start control) | stst17 | I'm able to continue working even when severely tired, if something really needs to be done.* | 0.11 | 0.40 | 0.19 |

*Note.* BFI-K: short version of the Big Five Inventory; BAS: The Behavioral Activation Scale; ERQ: Emotion Regulation Questionnaire; BIMS: The Barrat Impulsiveness Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed.

**Table S18**

*EFA results with three factors of the 14 items identified as the best predictors of the success of persistence conflicts*

| Scale (subscale) | Name | Item text | Factor loadings | | | $r_{outcome}$ |
|---|---|---|---|---|---|---|
| | | | Self-discipline | Positive outcome expectation | Meta-cognition | |
| BFI-K (neuroticism) | bfi11 | **tends to feel depressed, blue.** | -0.25 | **-0.40** | | -0.23 |
| BAS (reward responsiveness) | bisbas7 | **When I get something I want, I feel excited and energized.** | | **0.29** | | 0.15 |
| CSCS (persistence) | hoyle20 | I find it easy to keep with good behavior.* | 0.31 | 0.11 | 0.23 | 0.20 |
| IPIP (competence) | ipc1_4 | **Don't understand things. (R)** | | **0.65** | | 0.20 |
| IPIP (competence) | ipc1_8 | Complete tasks successfully.* | 0.29 | 0.36 | | 0.21 |
| IPIP (self-discipline) | ipc5_2 | **Find it difficult to get down to work.** | **0.70** | | | 0.26 |
| IPS | ips5 | **At the end of the day, I know I could have spent the time better.** | **-0.68** | | | -0.25 |
| MISCS (metacognitive knowledge) | MISCS12 | **I use different regulatory strategies depending on the situation.** | | | **0.56** | 0.20 |
| MISCS (metacognitive regulation) | MISCS15 | **I think of several strategies to resolve a self-control conflict and choose the best one.** | 0.18 | -0.13 | **0.50** | 0.18 |
| MISCS (metacognitive knowlede) | MISCS4 | **I understand my strengths and weaknesses when dealing with self-control conflicts.** | | 0.26 | **0.42** | 0.22 |
| SCS | scs06 | **I wish I had more self-discipline. (R)** | **0.74** | | | 0.23 |
| STSC (start control) | stst12 | **I find it difficult having to restart something after I thought I was already done. (R)** | **0.38** | | | 0.20 |
| STSC (start control) | stst16 | When my mind wanders while I'm reading, It's easy for my to concentrate on the text again.* | 0.16 | 0.28 | 0.16 | 0.21 |
| STSC (start control) | stst17 | I'm able to continue working even when severely tired, if something really needs to be done.* | | 0.30 | 0.28 | 0.19 |

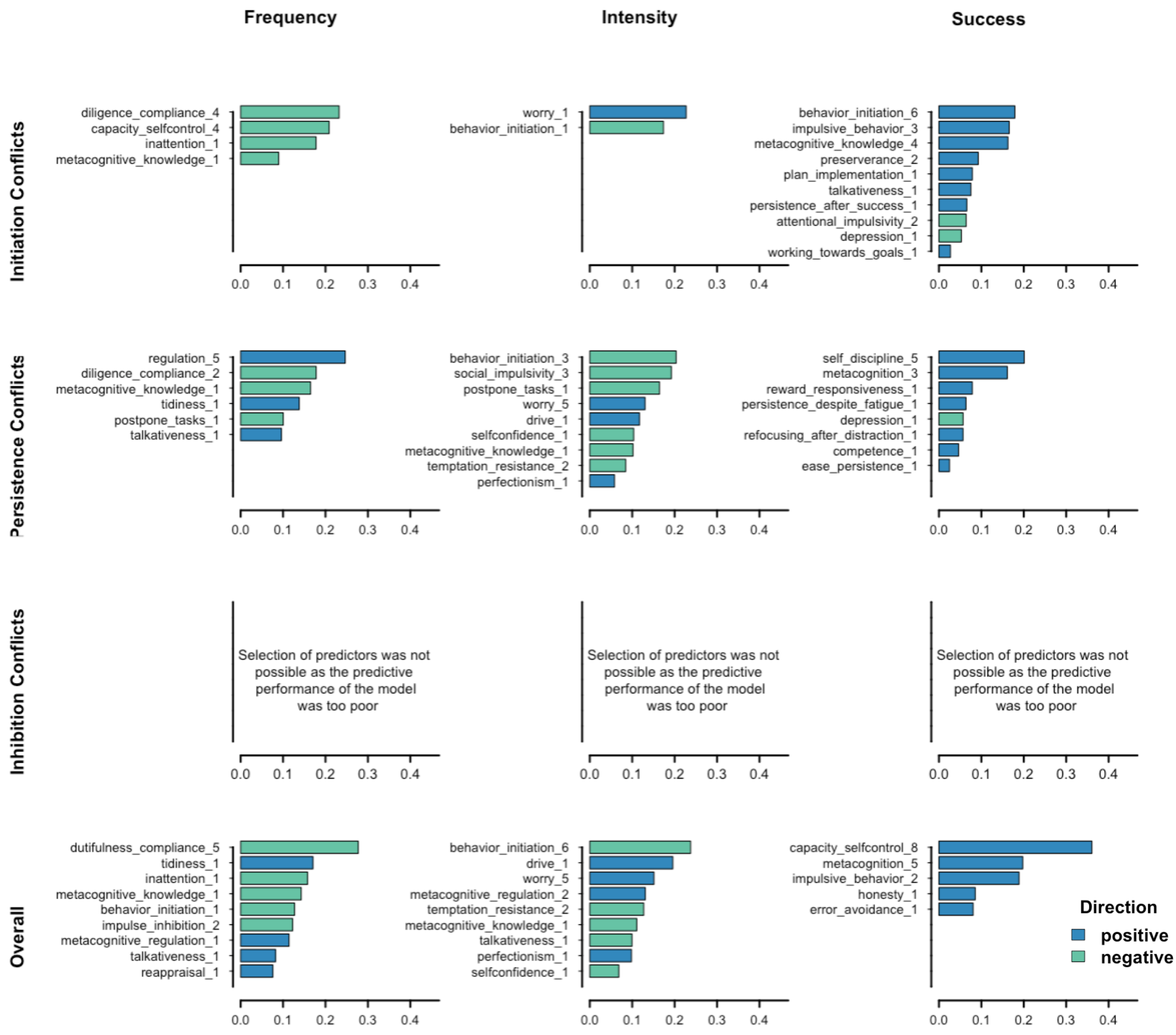
*Note.* BFI-K: short version of the Big Five Inventory; BAS: The Behavioral Activation Scale; ERQ: Emotion Regulation Questionnaire; BIMS: The Barrat Impulsiveness Scale; CSCS: The Capacity for Self-Control Scale; IPIP: International Personality Item Pool; MISCS: Metacognition in Self-Control Scale; SCS: The Self-Control Scale. Items and factor loadings highlighted in bold indicate that we would assign that item to the composite score of the factor; * indicates that we would keep the item separate in further analysis; R indicates that the item was reverse scored before entering the EFA. Loadings < |.10| are not displayed

## Standardized partial regression coefficients of the most important trait predictors for each outcome

We preregistered that we would base the comparison of the coefficients on partial regression coefficients and only later noticed that these may not be ideally suited for our research goals. Nevertheless, for the sake of transparency, Figure S10 represents the results of the partial regression models-

**Figure S1**

*Standardized partial regression coefficients of the most important trait predictors for each outcome*




*Note.* The number at the end of the name of the trait predictor denotes the number of items the predictor is composed of.

## ABCD Coding

To examine the extent that the trait items reflect affect (A), behavior (B), cognition (C), or desire (D), the items were coded as described in Wilt and Revelle (2015, p. 481 – 482). However, the instruction presented in Wilt and Revelle was extended to ask raters to focus explicitly on the wording of the item, addressing ambiguities in the instruction revealed during an initial pilot test of the coding procedure. The extended instruction can be found in the OSF project. Items were presented in a random order and coded by five student assistants and the first author.

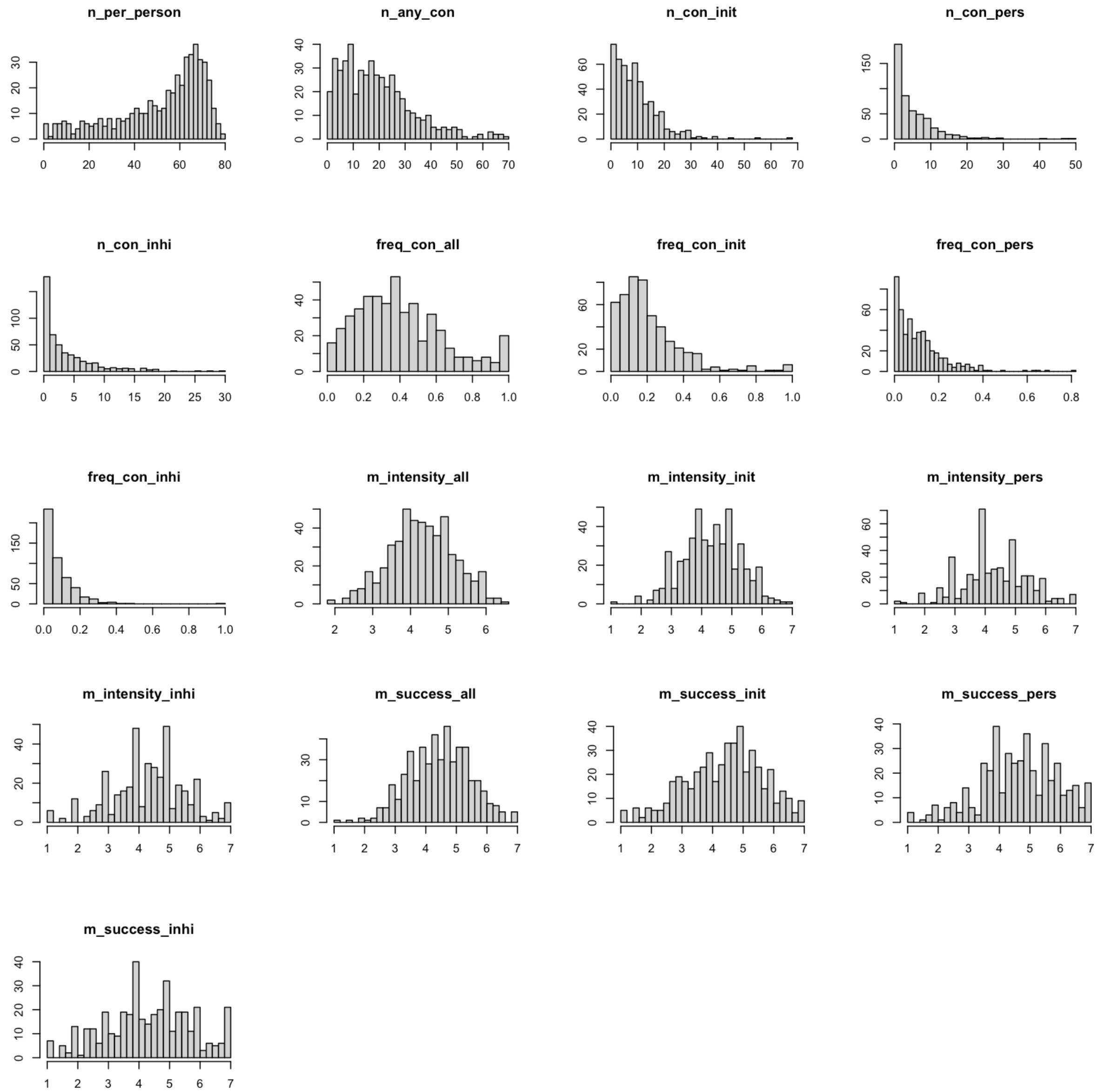
Since only the first author had been familiar with the coding scheme before, the six raters first rated a test set containing 20 items that were not part of the current study. The test items were taken from five different scales and can be found in the OSF project. ICCs for a single rater (ICC11) were .63 (affect), .69 (behavior), .54 (cognition), and .42 (desire). ICCs as the average of six random raters (ICC16) were .91 (affect), .93 (behavior), .87 (cognition), and .81 (desire). For the current project, the ICC16s are the most relevant reliability estimate as only the average coding across all raters would be used. Since the ICC16s were above .80 for all domains, we concluded that the raters understood the written instruction well and proceeded directly with the main coding without any verbal communication about the coding instruction.

In the main coding, ICCs for a single rater (ICC11) were .54 (affect), .45 (behavior), .51 (cognition), and .25 (desire). ICCs as the average of six random raters (ICC16) were .87 (affect), .83 (behavior), .86 (cognition), and .67 (desire). The estimates for the ICC16s were very similar to those of Big Five items reported in Wilt and Revelle (2015), which were .88 (affect), .83 (behavior), .88 (cognition), and .78 (desire). Although both studies suggest that the codings for desire seem to vary most across raters, the absolute reliability estimate for desire was slightly higher in Wilt and Revelle (2015) than in the current study. This difference may be explained by the finding that the items in our study had on average more desire content than the Big Five items which scored mostly very low on desire. For further analyses, the codings of each item were averaged across all items.

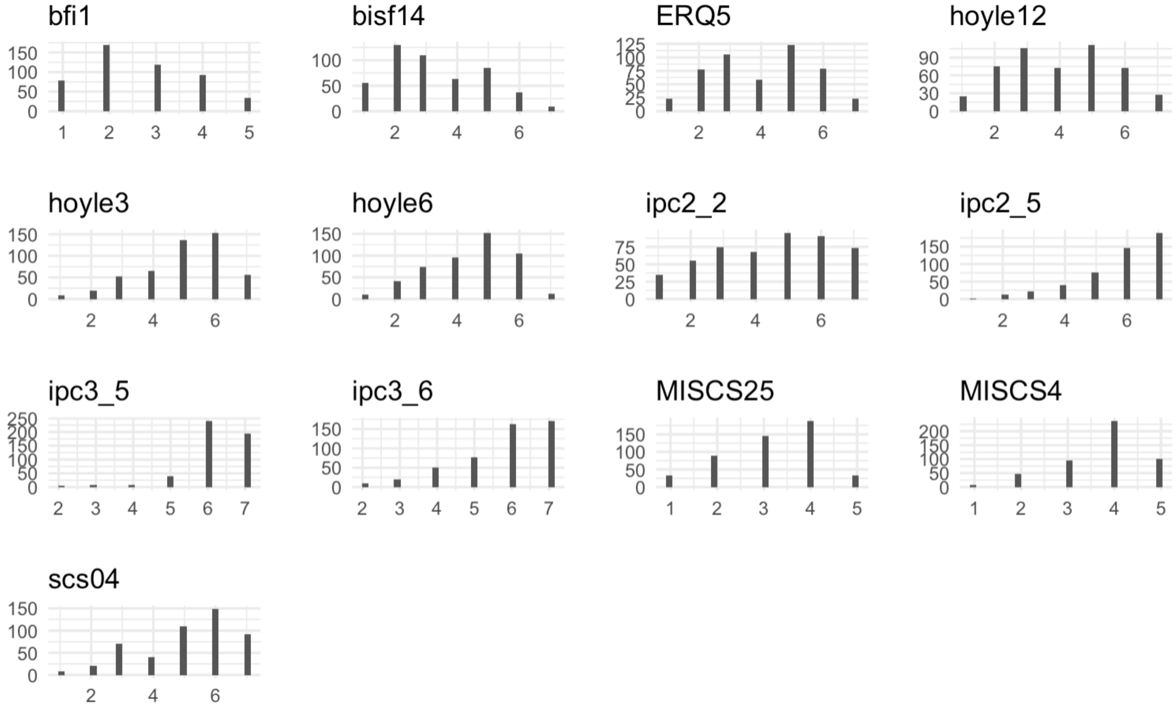
The results of the ABCD ratings of the selected trait predictors can be found in the result folder in the github repository with the name ‘ABCD results’.

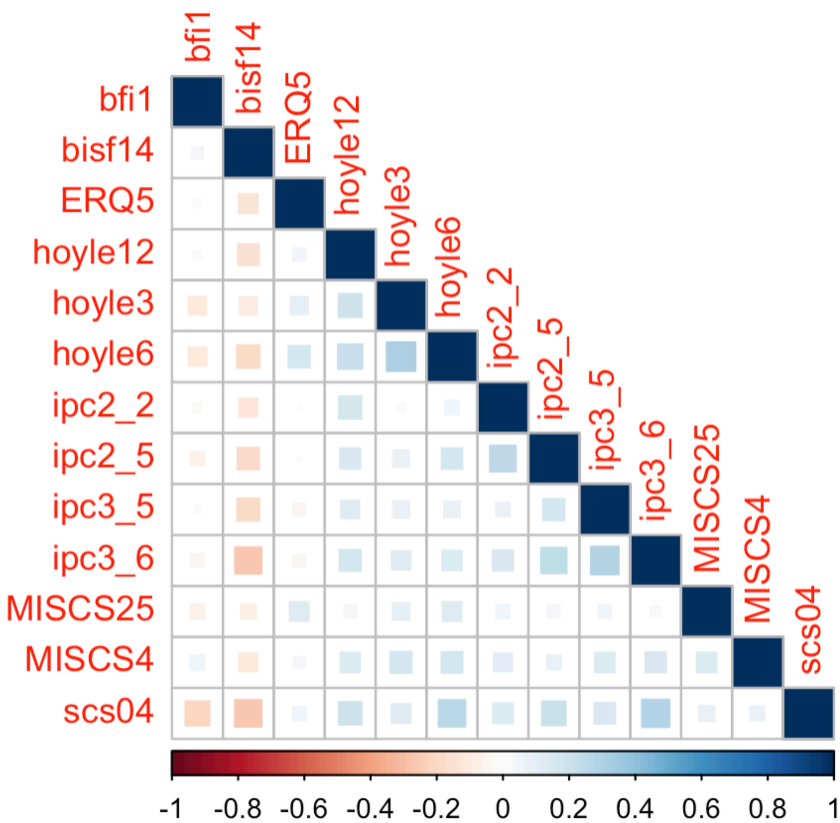
# Descriptive statistics

## Outcome variables

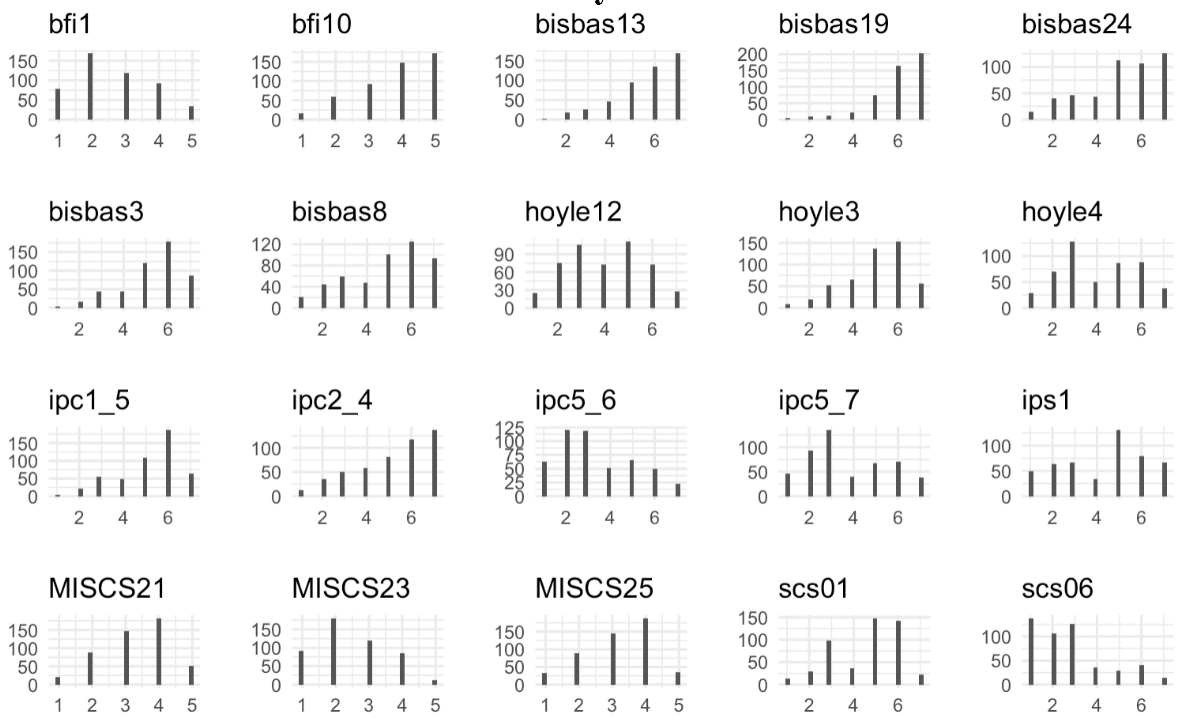


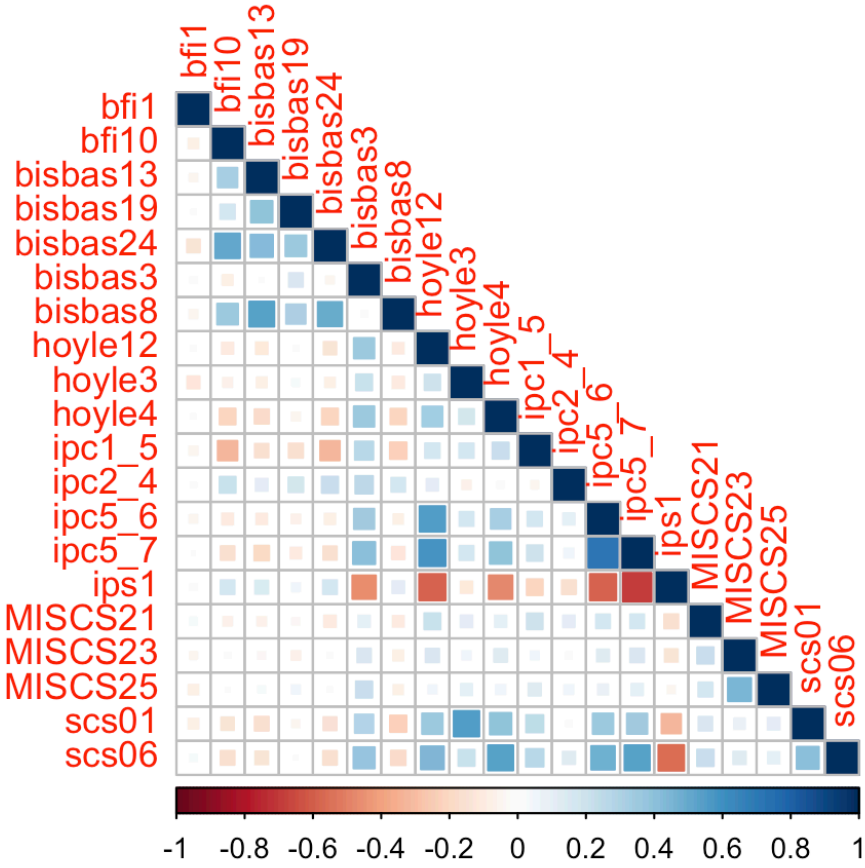
**Predictor items selected for the frequency across conflicts**



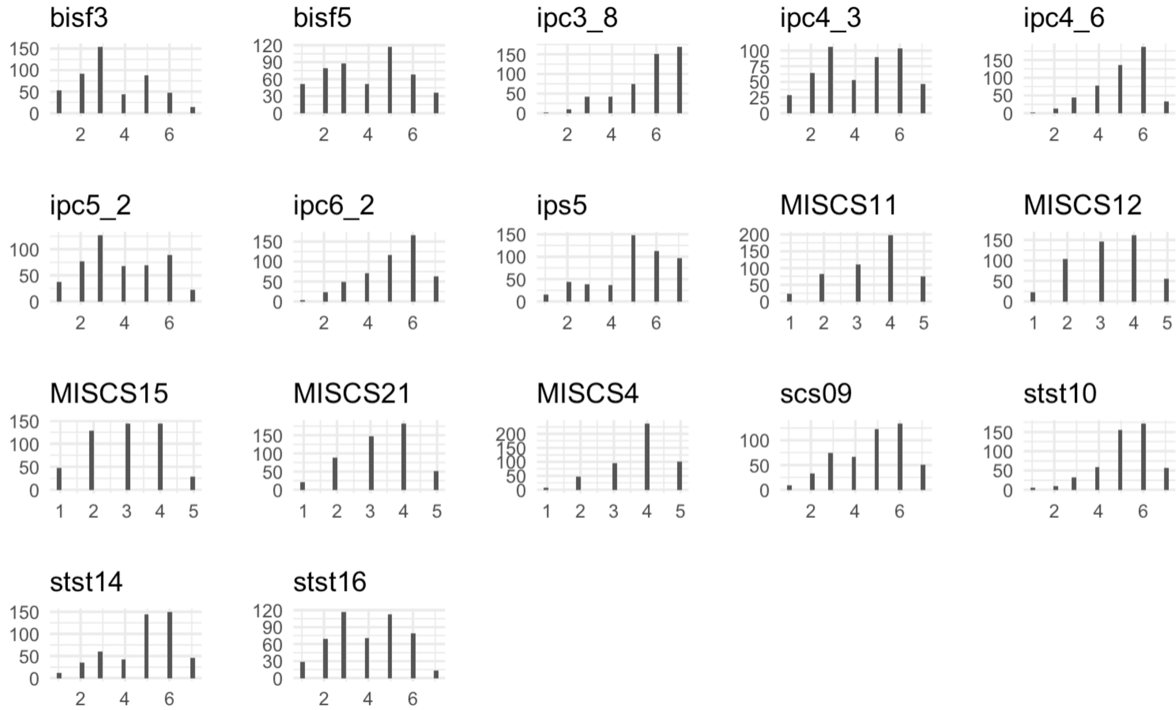



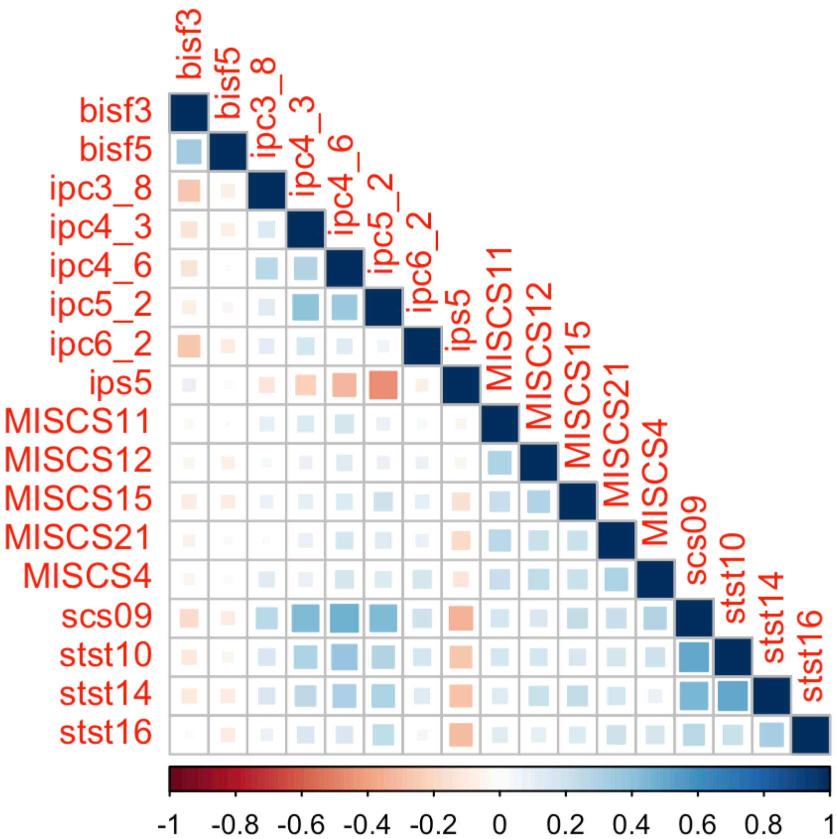
**Predictor items selected for the intensity across conflicts**



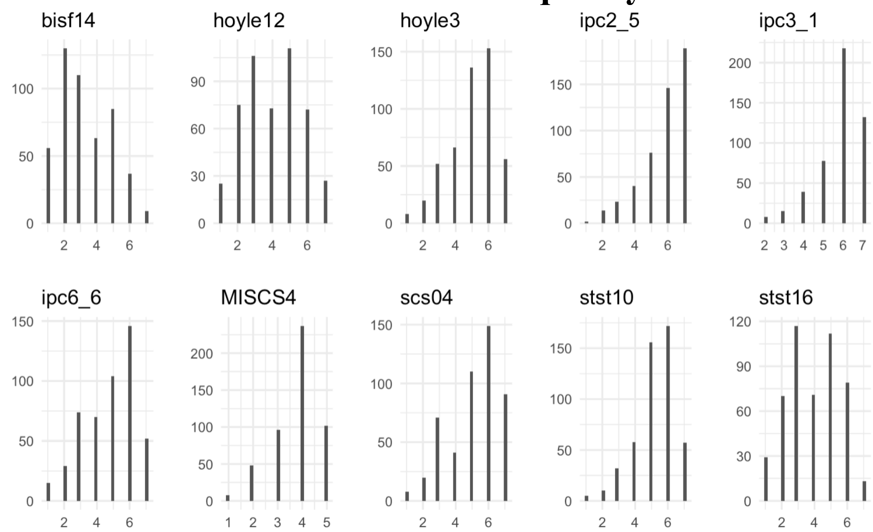



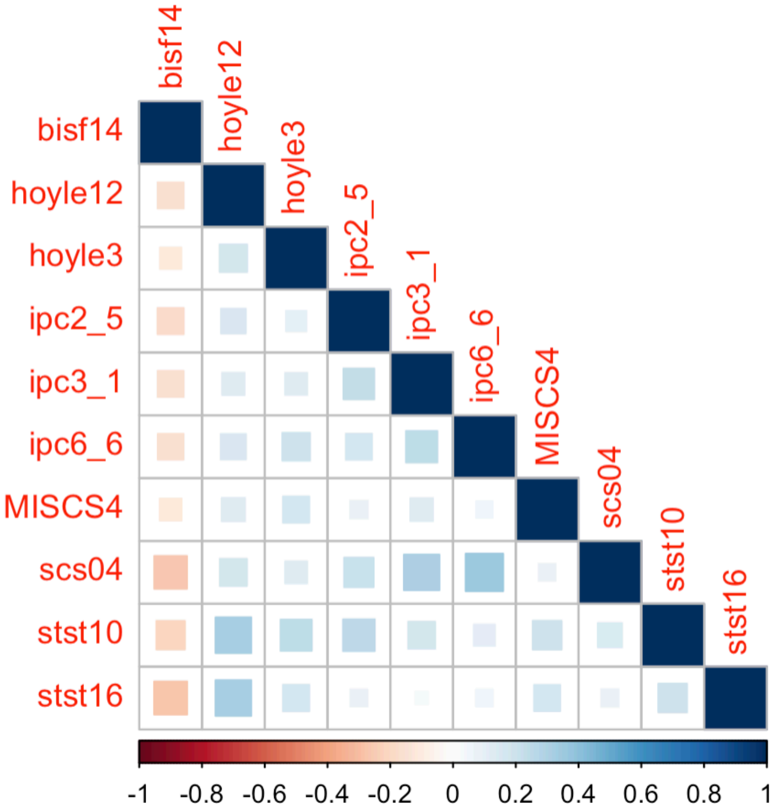
## Predictor items selected for the success across conflicts



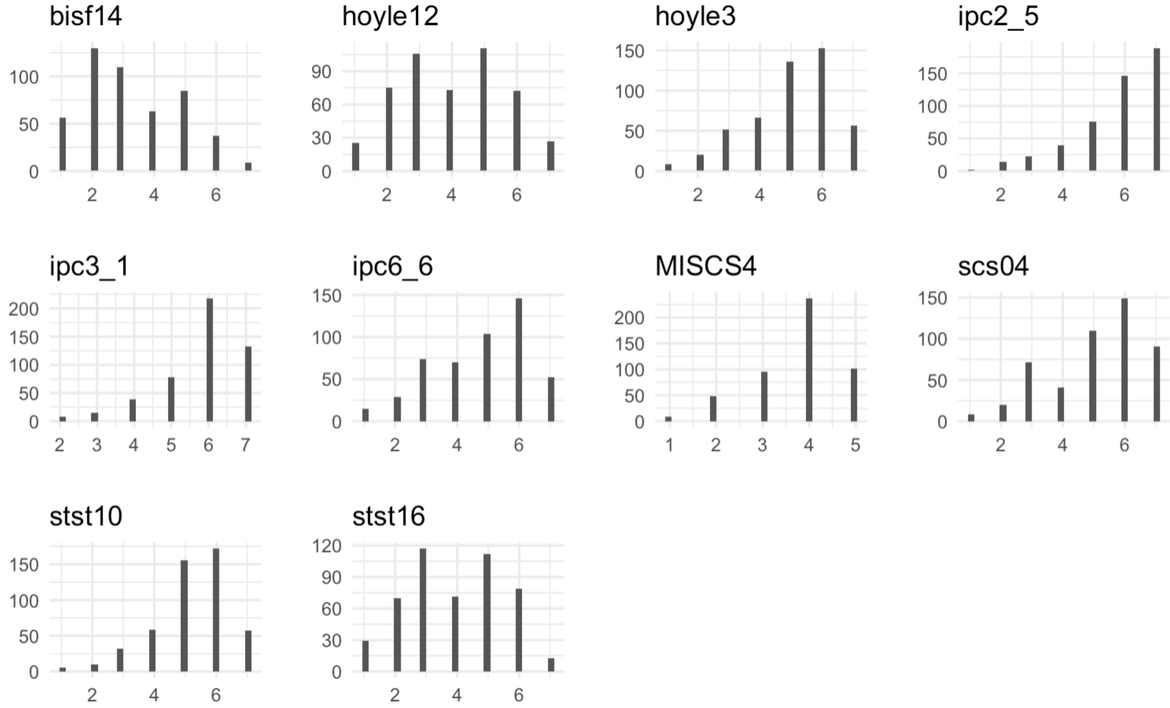



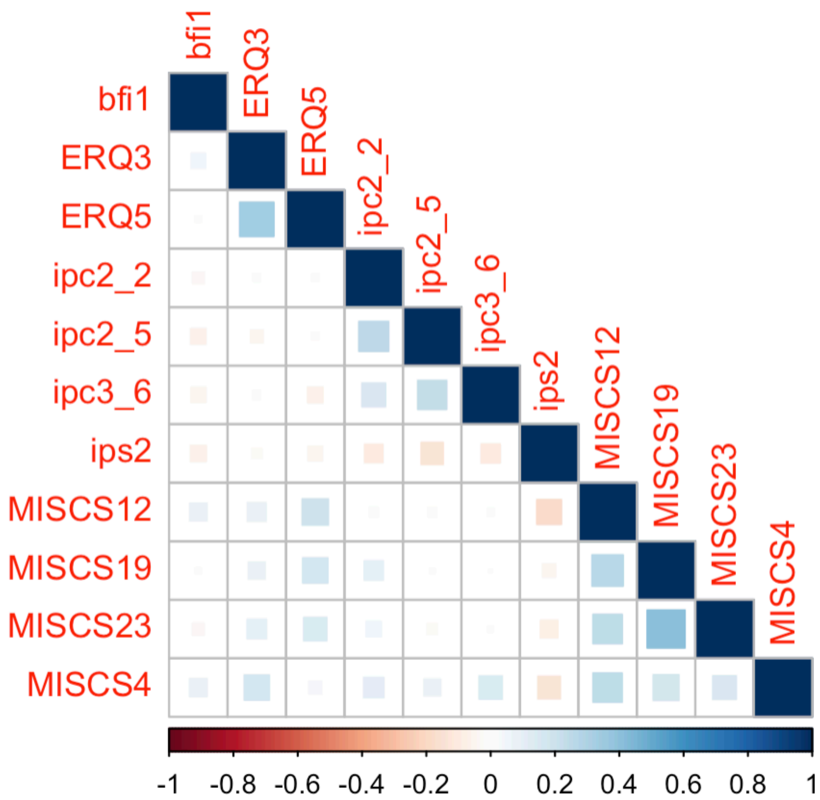
**Predictor items selected for the frequency of initiation conflicts**



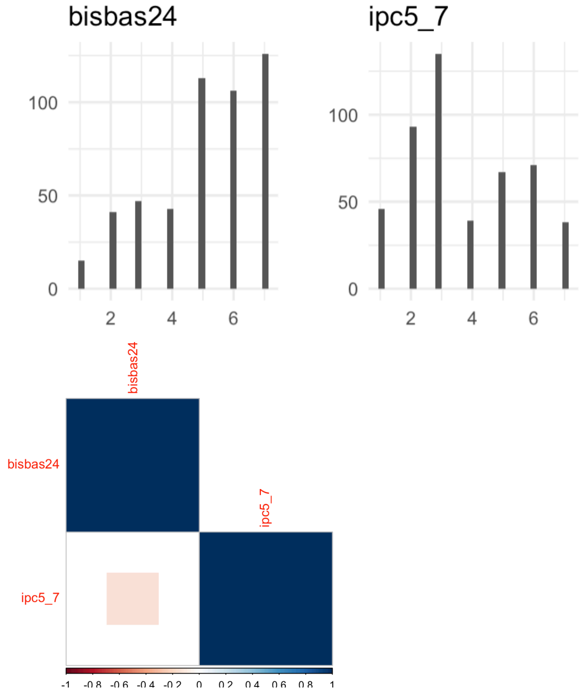



**Predictor items selected for the frequency of persistence conflicts**

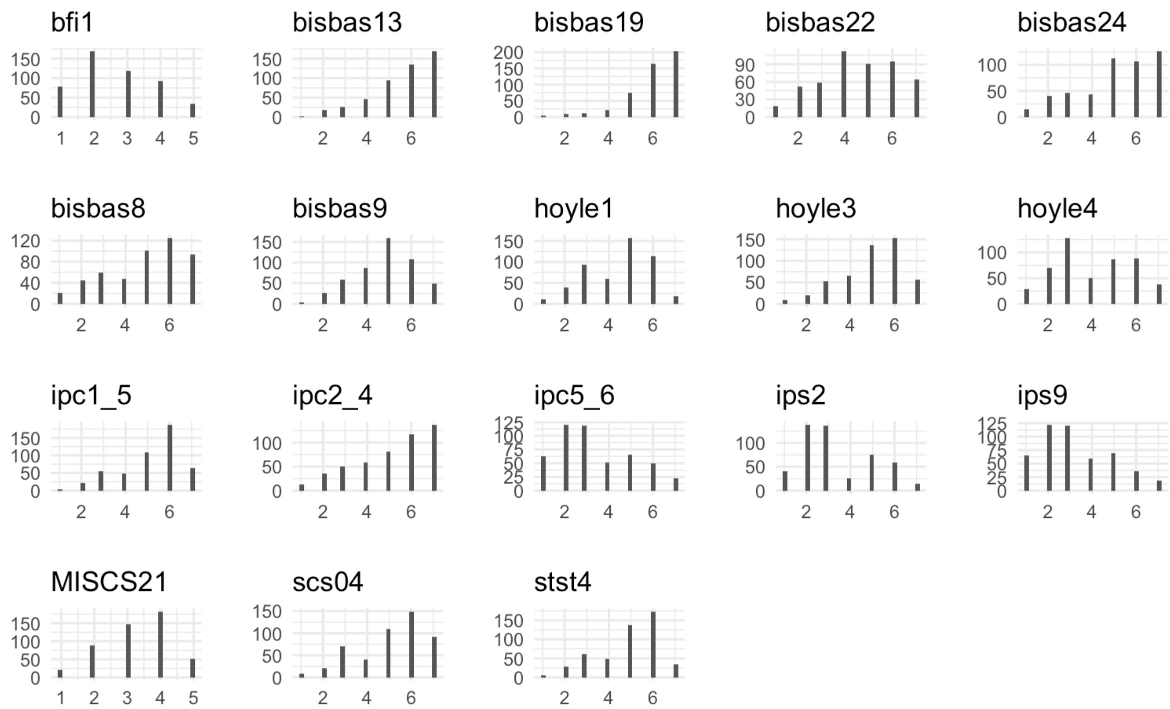


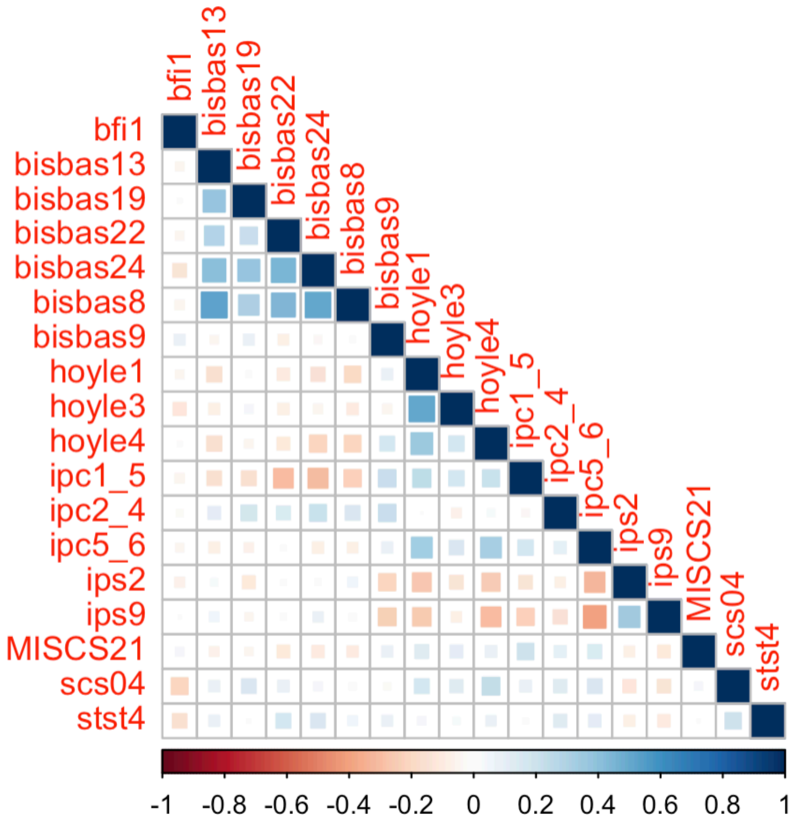



## Predictor items selected for the intensity of initiation conflicts

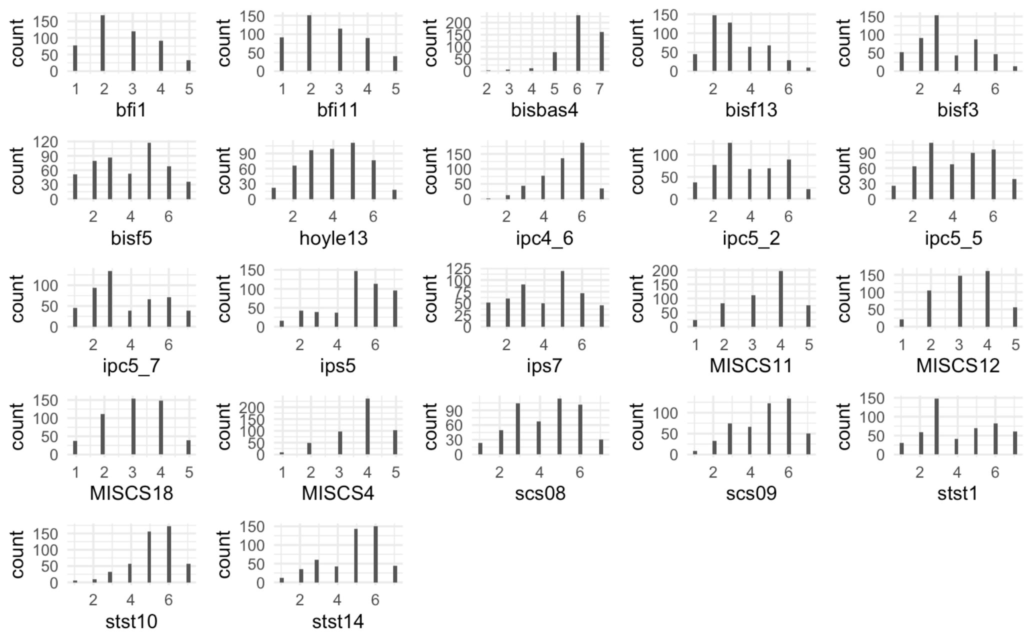


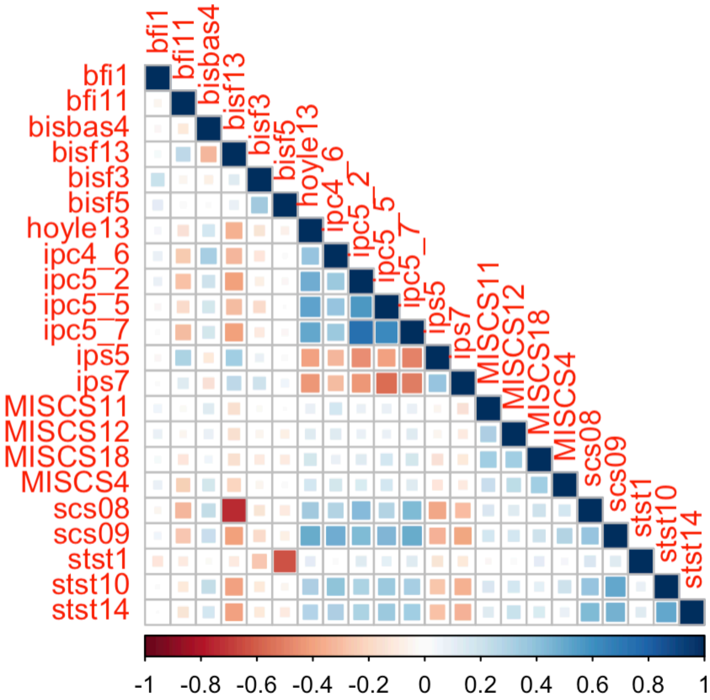

## Predictor items selected for the intensity of persistence conflicts



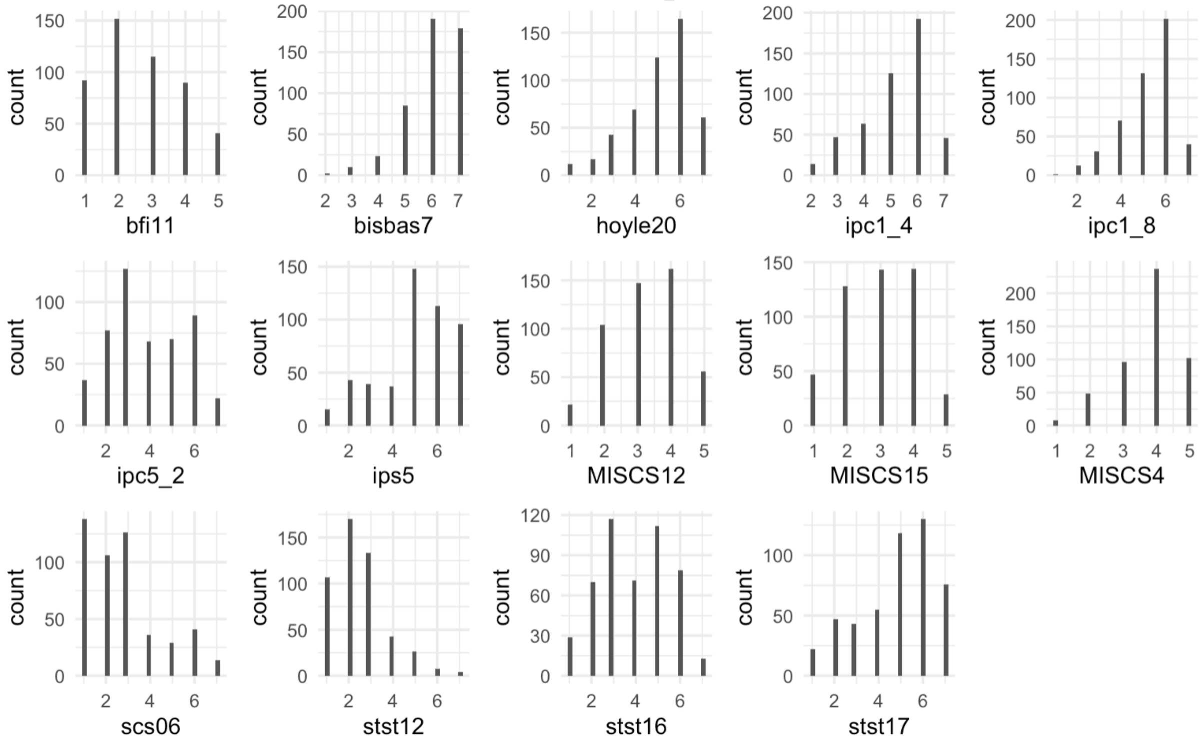


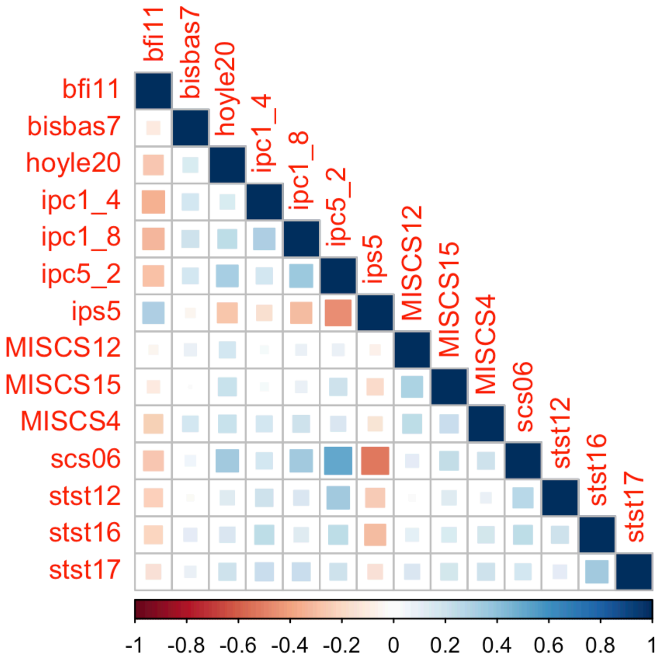

**Predictor items selected for the success of initiation conflicts**






**Predictor items selected for the success of persistence conflicts**




bfi11
bisbas7
hoyle20
ipc1_4
ipc1_8
ipc5_2
ips5
MISCS12
MISCS15
MISCS4
scs06
stst12
stst16
stst17
-1 -0.8 -0.6 -0.4 -0.2 0 0.2 0.4 0.6 0.8 1